

Higher-Order Asymptotics of Regression-Adjusted Estimators in Randomized Trials*

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Abstract

Regression adjustment is widely used to estimate average treatment effects in randomized trials. Under balanced assignment, simple regression (REG, ANCOVA) and interacted regression (IREG, ANCOVA II) have the same first-order variance, leaving their relative performance to higher-order terms. We derive a closed-form expression for the difference in their $1/n^2$ mean-squared-error terms in a fixed-dimensional superpopulation framework under complete randomization. The difference depends jointly on covariate shape, treatment-effect heterogeneity, residual heteroskedasticity, and nonlinear misspecification, and can favor either estimator. In one-dimensional designs with skewed covariates, the resulting asymptotic deficiency ranges from a few observations to more than one hundred. We further study practical inference, showing that without treatment-effect heterogeneity, the common pairing of IREG with HC1 standard errors can undercover substantially in skewed, heteroskedastic DGPs. HC3 largely improves coverage but does not fully resolve the most extreme leverage configurations. For inference on the population ATE under superpopulation sampling, we also incorporate the known superpopulation variance correction for IREG; this correction becomes important as treatment-effect heterogeneity grows. However, when heterogeneity is small it artificially widens intervals, making REG with HC3 the more efficient combination in our low-heterogeneity designs.

1 Introduction

Randomized controlled trials (RCTs) are fundamental to how causal effects are ascertained across a number of fields. One valuable attribute of RCTs is that they enable unbiased estimation of average treatment effects using simple methods such as the difference in means, without reliance on parametric assumptions [Imbens and Rubin, 2015]. However, while the difference-in-means estimator is unbiased, this does not automatically make it the preferred approach; and in particular regression adjustments are often used to improve power. For example, recent guidance from the U.S. Food and Drug Administration encourages adjustment for baseline covariates that are prognostic for the outcome [U.S. Food and Drug Administration, 2023].

The study of regression adjustments in RCTs has a long tradition in statistics. Fisher [1935] used arguments rooted in parametric linear modeling to advocate for such regression

*Code, simulation scripts and the computational certificates used in the proofs: <https://github.com/kelleyan-pixel/higher-order-regression-adjustment-for-rct>.

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adjustments. Fisher’s analysis was later critiqued by [Freedman \[2008\]](#), who argued that reliance on linear models was misaligned with model-free guarantees sought in RCT inference. [Lin \[2013\]](#), however, responded to Freedman’s critique by showing that regression adjustments do not require parametric assumptions to be near-unbiased for the average treatment effect. Furthermore, [Lin \[2013\]](#) showed that regression adjustments with full treatment-control interaction weakly dominate both the difference in means estimator and Fisher’s simple regression adjustment in terms of their leading-order (root- n scale) errors. Lin’s weak dominance result has led such treatment-interacted regressions to be often presented as a safe starting point for use of regression adjustments in RCTs [[Li and Ding, 2017](#), [Negi and Wooldridge, 2021](#), [Wager, 2025](#)].

A gap in this narrative, however, is that the first-order advantage of treatment-interacted regression over simple regression is strict only when the randomization probability satisfies $\pi \neq 0.5$ (and treatment effects vary with the covariates); the dominance is weak at every π . In evenly randomized trials with $\pi = 0.5$, the first-order accuracy of both methods is equal for any data-generating distribution, i.e., first-order theory offers no guidance on which method to prefer. And, while $\pi = 0.5$ is of course a special case, this is an important special case in applications, and it is not a priori clear we should prefer interacted regression to simple regression here. For example, [Zhao and Ding \[2023\]](#) emphasize the finite-sample trade-off associated with the larger number of parameters in interacted regression, arguing that simple regression may be more stable with limited sample size—especially when there are multiple treatment levels.

To see what may be at stake in practice, consider an example modeled after a typical tech industry setting. Say a company runs an A/B test on a sample of its user base and adjusts for X_i , each user’s revenue or spend in the prior period. Such measures are strongly right-skewed because a small number of heavy users account for much of the total. Suppose also that these users have more volatile outcomes, so that outcome noise increases with X_i , while the treatment shifts every user’s outcome by the same amount. Specifically, let the raw X_i be marginally $\text{LogNormal}(0, 1.1^2)$ and let

$$\sigma(X_i) = 1 + X_i, \quad Y_i(0) = 0.9 X_i + \sigma(X_i)u_i, \quad Y_i(1) = Y_i(0) + 1,$$

with $u_i \stackrel{\text{iid}}{\sim} N(0, 1)$. There is no treatment-effect heterogeneity, so the heterogeneity share R_τ^2 of Section 2 is zero, and the whitened covariate has skewness 8.2. In simulation, interacted regression MSE exceeds that of simple regression by 6.7% at $n = 500$, and asymptotically, the former costs around 83 users’ worth of precision. Furthermore, inference using interacted regression is sensitive to the choice of standard error estimator. At $n = 500$ and level $\alpha = 0.05$, HC1 undercovers at 93.5% (fixed by switching to HC3); meanwhile, simple regression is well-calibrated with both error estimates.

Motivated by this gap, we conduct a higher-order asymptotic analysis of both simple and interacted regression adjustments in RCTs to elucidate the relative merits of the two approaches when $\pi = 0.5$. We frame our findings in terms of asymptotic deficiency [[Hodges and Lehmann, 1970](#)]. When two estimators have matching first-order behavior but one dominates the other on the second-order scale then, asymptotically, switching to the better one gives an accuracy boost that is equivalent to a fixed boost in sample size, which is referred to as the deficiency. Our simulations include settings with moderate misspecification or heteroskedasticity in which the second-order comparison favors simple regression. Despite this tendency, our results show that there is no universal second-order ordering. For instance, in a well-specified model with Gaussian covariates, interacted regression’s advantage actually grows linearly with the number of regressors. This highlights that finite-sample performance depends not only on model dimension but also on higher-order features of the covariate and outcome distributions.

As a practical companion to this theory, we study confidence intervals from standard implementations in the R package `estimatr`: `lm_lin` for interacted regression and `lm_robust` for simple regression. Because this software is meant for a fixed-design framework and thus omits an additional G/n component of superpopulation uncertainty (G is the variance of the linear part of the treatment effect, defined in Section 3.2), we examine both settings with $G = 0$, which isolates finite-sample variance-estimation behavior, and settings with moderate to large treatment-effect heterogeneity, which illustrates effects from both estimator choice and superpopulation randomness. With $G = 0$, interacted regression with HC1 can undercover substantially under skewed, tail-heteroskedastic covariates; HC3 improves coverage substantially in most designs but remains imperfect under extreme tails and at leverage-one observations. Simple regression exhibits much milder HC1 distortions, while HC3 generally brings coverage closer to nominal. When heterogeneity is large, the omitted G/n component becomes important: the conventional standard errors undercover substantially, while the feasible correction closely restores nominal coverage. When heterogeneity is small or absent, by contrast, the plug-in boosts coverage but is inefficient and miscalibrated across nominal levels. Thus, interacted regression introduces two additional inferential complications in the superpopulation setting: its centered implementation requires accounting for an estimand-level variance component, while its arm-specific fit can make finite-sample variance estimation more sensitive to skewness and leverage in the covariates.

1.1 Related Work

Regression adjustment in randomized experiments. The literature on regression adjustment in randomized experiments spans two complementary frameworks. The design-based literature treats the potential outcomes and covariates as fixed in a finite population, with randomness arising from treatment assignment. This is the framework of Lin [2013], Li and Ding [2017], and Zhao and Ding [2023]. A complementary superpopulation framework treats the experimental units themselves as random draws and studies repeated sampling over both the units and treatment assignment. Yang and Tsiatis [2001], Negi and Wooldridge [2021], Ye et al. [2023] and Wager [2025] work in this setting. Our analysis follows the latter framework. This distinction is important because our goal is to characterize higher-order sampling behavior of REG and IREG under IID sampling rather than finite-population randomization error; as Section 3 shows, it also determines which variance estimator is appropriate.

The relative performance of REG and IREG has been studied primarily through first-order asymptotic efficiency. Yang and Tsiatis [2001] compare ANCOVA with and without treatment interactions in a superpopulation pretest-posttest setting, and Lin [2013] shows that IREG weakly dominates REG and the difference-in-means estimator in leading-order asymptotic variance for general treatment probabilities, with equality between REG and IREG under balanced treatment assignment. Zhao and Ding [2023] study the finite-sample consequences of the larger interacted specification and emphasize the role of the additional parameters introduced by treatment-covariate interactions, particularly in settings with multiple treatment levels. Our results add nuance to this perspective: in the balanced Gaussian, homoskedastic, well-specified case the second-order coefficient favors IREG and, with p fixed as $n \rightarrow \infty$, is linear in p , while outside this setting the sign of the second-order comparison depends on skewness, tail behavior, heteroskedasticity and misspecification.

Higher-order asymptotic theory. First-order asymptotic approximations cannot distinguish estimators with the same limiting distribution, even when their finite-sample behavior differs substantially. Higher-order asymptotic theory addresses this through Edgeworth expan-

sions and the associated moment (Nagar-type) expansions; see, for example, [Sargan \[1974\]](#), [Bhattacharya and Rao \[1976\]](#), [Hall \[1992\]](#) and [Ghosh \[1994\]](#). [Hodges and Lehmann \[1970\]](#) introduced asymptotic deficiency as a measure of the additional sample size required by a less accurate estimator, while [Rothenberg \[1984\]](#) emphasized the role of higher-order approximations when competing estimators share the same limiting distribution. Parallel to our contribution is [Newey and Smith \[2004\]](#), which includes higher-order comparisons of first-order equivalent GMM and generalized empirical likelihood estimators.

Recent work also studies higher-order regression adjustment under a design-based framework with many covariates. [Lei and Ding \[2021\]](#) characterize the bias of Lin’s estimator when p grows with n and propose a bias-corrected estimator, and [Chiang et al. \[2025\]](#), in Neyman’s design-based framework with the number of covariates growing more slowly than n , derive the first- and second-order terms of the stochastic expansions of regression-adjusted estimators, analyze the higher-order behavior of existing inference procedures, and propose a cross-fitted estimator and a modified HC3 standard error. Related work studies debiased and exactly bias-corrected adjustment [[Lu et al., 2025](#), [Chang et al., 2024](#)], many regressors under covariate-adaptive randomization [[Jiang et al., 2025](#)], and estimators motivated by higher-order influence functions [[Zhao et al., 2024](#)]. Closest in spirit to our analysis is [Tan \[2014\]](#), who compares first-order-equivalent regression and likelihood (calibration) estimators at second order in sampling and missing-data problems with a fixed number of auxiliary variables; the likelihood estimators are second-order unbiased but the regression estimators are not, and no estimator studied always has the smallest second-order variance, even after bias correction. Our comparison differs in the pair of estimators (pooled versus interacted linear working models), in the target (an average treatment effect) and in the design (complete randomization).

Variance estimation. The important fact that inference for a population (rather than sample) average treatment effect after interacted regression requires an additional variance component is discussed by [Imbens and Wooldridge \[2009\]](#), [Negi and Wooldridge \[2021\]](#), [Zhao and Ding \[2021\]](#) and [Ye et al. \[2023\]](#). [Samii and Aronow \[2012\]](#) establish the equivalence between design-based Neyman variance estimators and heteroskedasticity-robust regression variances; `estimatr` uses HC2 by default. Finite-sample behavior of heteroskedasticity-robust variance estimators is studied by [Chesher and Jewitt \[1987\]](#), [MacKinnon and White \[1985\]](#), [Long and Ervin \[2000\]](#) and [Imbens and Kolesár \[2016\]](#); [Lei and Ding \[2021\]](#) recommend HC3-type corrections for regression adjustment with many covariates.

Estimands. Our expansion concerns the population ATE under IID superpopulation sampling. Much of the design-based literature, including [Lin \[2013\]](#), [Lei and Ding \[2021\]](#) and [Chiang et al. \[2025\]](#), targets the sample ATE of a fixed finite population, for which first-order variances differ and higher-order constants generally differ as well. Our own comparison of complete randomization with Bernoulli assignment shows how sensitive the n^{-2} constant is to the probabilistic framework (Theorem 10 shifts it by $4G$), so the constants of Theorem 2 should not be carried over to finite-population analyses without recomputation. The superpopulation view also matters for inference: the covariate mean at which IREG is centered is itself random, which is the source of the variance component discussed in Section 3.2.

2 Second-Order Analysis

We focus on completely randomized trials (CRTs) with causal effects formalized using the potential outcomes model [[Imbens and Rubin, 2015](#)]. We assume the covariates $X_i \in \mathbb{R}^p$

and potential outcomes $Y_i(0), Y_i(1) \in \mathbb{R}$ are drawn IID from a distribution P , with average treatment effect $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$. Treatment is assigned randomly conditional on the fixed treated group size $n_1 = \sum_{i=1}^n W_i = \lfloor \pi n \rfloor$; treatment assignments are exchangeable but not independent. Because both regression estimators include an intercept, translating the covariates does not change the estimators, so we work without loss of generality with centered covariates, $\mathbb{E}[X] = 0$.

The analyst observes (X_i, Y_i, W_i) for $i = 1, \dots, n$, where $Y_i = Y_i(W_i)$. We consider the following three estimators of the ATE:

$$\begin{aligned}\hat{\tau}_{DM} &= \frac{1}{|\{i : W_i = 1\}|} \sum_{\{i : W_i = 1\}} Y_i - \frac{1}{|\{i : W_i = 0\}|} \sum_{\{i : W_i = 0\}} Y_i, \\ \hat{\tau}_{REG} &= \hat{\tau}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau, \\ \hat{\tau}_{IREG} &= \hat{\tau} + \frac{1}{n} \sum_{i=1}^n X_i \hat{\delta}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau + W_i X_i \delta,\end{aligned}\tag{1}$$

where, e.g., $Y_i \sim \alpha + X_i\beta + W_i\tau$ means we algorithmically run a regression of Y_i against X_i and W_i with an intercept (without making any claims on well-specification of such a regression). To describe their first-order behavior it is helpful to define the best linear approximation to the conditional response function of the potential outcomes,

$$\left\{ \alpha_{(w)}^*, \beta_{(w)}^* \right\} = \underset{\alpha_{(w)}, \beta_{(w)}}{\operatorname{argmin}} \left\{ \mathbb{E} \left[(Y_i(w) - X_i\beta_{(w)} - \alpha_{(w)})^2 \right] \right\},\tag{2}$$

together with the oracle residuals and the scaled heterogeneity direction

$$\begin{aligned}\varepsilon_i(w) &= Y_i(w) - \alpha_{(w)}^* - X_i^\top \beta_{(w)}^*, \\ Z_i &= \Sigma^{-1/2} X_i, \quad \Sigma = \operatorname{Var}[X_i], \quad \gamma = \Sigma^{1/2} (\beta_{(1)}^* - \beta_{(0)}^*),\end{aligned}\tag{3}$$

so that $\mathbb{E}[Z_i] = 0$, $\mathbb{E}[Z_i Z_i^\top] = I_p$, $\mathbb{E}[Z_i \varepsilon_i(w)] = 0$ and $\varepsilon_i = \varepsilon_i(W_i)$. All population quantities below are invariant to the choice of matrix square root: replacing $\Sigma^{1/2}$ by $\Sigma^{1/2} O$ for an orthogonal O maps $Z \mapsto O^\top Z$ and $\gamma \mapsto O^\top \gamma$, under which every scalar in Theorem 2 is unchanged. The first-order behavior of the three estimators is well known [Negi and Wooldridge, 2021].

Proposition 1. *Suppose that X_i and $Y_i(w)$ have finite second moments and that $\operatorname{Var}[X_i]$ is invertible. Then all three estimators above are asymptotically normal, $\sqrt{n}(\hat{\tau} - \tau) \Rightarrow \mathcal{N}(0, V)$, with limiting variances*

$$\begin{aligned}V_{DM} &= V_{IREG} + \operatorname{Var} \left[X_i \left(\sqrt{\frac{\pi}{1-\pi}} \beta_{(0)}^* + \sqrt{\frac{1-\pi}{\pi}} \beta_{(1)}^* \right) \right], \\ V_{REG} &= V_{IREG} + \frac{(1-2\pi)^2}{\pi(1-\pi)} \operatorname{Var} \left[X_i (\beta_{(1)}^* - \beta_{(0)}^*) \right], \\ V_{IREG} &= \operatorname{Var} \left[X_i (\beta_{(1)}^* - \beta_{(0)}^*) \right] + \sum_{w=0,1} \frac{1}{\mathbb{P}[W_i = w]} \mathbb{E} \left[(Y_i(w) - X_i \beta_{(w)}^* - \alpha_{(w)}^*)^2 \right].\end{aligned}\tag{4}$$

Thus $\hat{\tau}_{IREG}$ weakly dominates both $\hat{\tau}_{DM}$ and $\hat{\tau}_{REG}$ at every π ; its advantage over $\hat{\tau}_{REG}$ is strict for generic distributions P when $\pi \neq 0.5$, while at $\pi = 0.5$ the two regression estimators have the same asymptotic variance

$$V^* = \|\gamma\|^2 + 4\mathbb{E}[\varepsilon_i^2] = \|\gamma\|^2 + 2\mathbb{E}[\varepsilon_i(1)^2] + 2\mathbb{E}[\varepsilon_i(0)^2].\tag{5}$$

This is the setting we study. Because first-order theory is indifferent between the two methods, we turn to second-order asymptotics.

2.1 Deficiency and the expansion MSE

Suppose we compare two estimators $\hat{\tau}_A$ and $\hat{\tau}_B$ of a shared quantity τ whose mean-squared errors agree on the (first-order) $1/n$ but not on the (second-order) $1/n^2$ scale,

$$\text{MSE}(\hat{\tau}_A) = \frac{V}{n} + \frac{c_A}{n^2} + o\left(\frac{1}{n^2}\right), \quad \text{MSE}(\hat{\tau}_B) = \frac{V}{n} + \frac{c_B}{n^2} + o\left(\frac{1}{n^2}\right). \quad (6)$$

Hodges and Lehmann [1970] propose quantifying the difference in performance by the number of extra samples required to compensate for using the less accurate estimator: they solve for k_n such that $\hat{\tau}_B$ run on k_n samples matches the accuracy of $\hat{\tau}_A$ on n samples as closely as possible. When $\lim_{n \rightarrow \infty} (k_n - n) = \Delta \in \mathbb{R}$ exists, Δ is the asymptotic deficiency of $\hat{\tau}_B$ relative to $\hat{\tau}_A$. Under (6),

$$\Delta = (c_B - c_A) / V. \quad (7)$$

Both regression estimators involve inverse sample Gram matrices, so exact finite-sample moments require conditions controlling singularity and near-singularity. For maximum generality (for instance to distributions with atoms), we work with the second moment of the stochastic expansion rather than the exact MSE. Under the additional regularity conditions that the estimators are well defined and the Taylor remainder is negligible in mean square at the n^{-2} scale, the expansion MSE agrees with the usual finite-sample MSE up to $o(n^{-2})$ and can be considered equivalent.

Definition 1 (Expansion MSE). Let $\mathcal{V}_i$ collect the sample-moment coordinates entering the coefficient function of an estimator, let $\mu = \mathbb{E}[\mathcal{V}_i]$, $\bar{\mathcal{V}} = n^{-1} \sum_i \mathcal{V}_i$, and suppose $\hat{\tau} = g(\bar{\mathcal{V}})$ for a function g that is three times continuously differentiable in a neighborhood of μ with $g(\mu) = \tau$. Write $\zeta^{(n)} = \sqrt{n}(\bar{\mathcal{V}} - \mu)$ and let

$$\tilde{\tau}_n = \tau + \frac{1}{\sqrt{n}} a_i \zeta_i^{(n)} + \frac{1}{2n} H_{ij} \zeta_i^{(n)} \zeta_j^{(n)} + \frac{1}{6n^{3/2}} A_{3,ijk} \zeta_i^{(n)} \zeta_j^{(n)} \zeta_k^{(n)} \quad (8)$$

be the third-order Taylor polynomial of g at μ , with a , H and A_3 the first three derivative tensors of g at μ . The expansion MSE is $\text{MSE}^{\text{exp}}(\hat{\tau}) = \mathbb{E}[(\tilde{\tau}_n - \tau)^2]$.

We make the following assumption for this quantity to converge:

Assumption 1 (Sampling and moments). $(X_i, Y_i(0), Y_i(1))$ are IID from P , $\Sigma = \text{Var}[X_i] \succ 0$, p is fixed, $\pi = 1/2$ with $n_1 = n/2$ for even n (and $n_1 = \lfloor n/2 \rfloor$ otherwise), and

$$\mathbb{E}[\|X_i\|^{12}] + \sum_{w=0}^1 \mathbb{E}[|Y_i(w)|^{12}] < \infty.$$

Note that this implies $\mathbb{E}\|\mathcal{V}_i\|^6 < \infty$ by the Cauchy–Schwarz inequality.

2.2 The second-order difference

Theorem 2. *Let Assumption 1 hold, and define*

$$\begin{aligned} K &= \mathbb{E}[ZZ^\top \|Z\|^2], & S &= \mathbb{E}[Z\|Z\|^2], & S_\gamma &= \mathbb{E}[ZZ^\top (Z^\top \gamma)], \\ \mathcal{E} &= \mathbb{E}[ZZ^\top \varepsilon], & \mathcal{E}_W &= \mathbb{E}[ZZ^\top (W - \tfrac{1}{2})\varepsilon] = \tfrac{1}{4}(\mathbb{E}[ZZ^\top \varepsilon(1)] - \mathbb{E}[ZZ^\top \varepsilon(0)]). \end{aligned}$$

With MSE^{exp} as in Definition 1 and W assigned via balanced CRT,

$$\text{MSE}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_p}{n^2} + O(n^{-3}),$$

where

$$\begin{aligned} D_p = & - (2p + 3)\|\gamma\|^2 + \gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2 \\ & - 16 \text{tr}(S_\gamma \mathcal{E}_W) - 8(\gamma^\top S) \text{tr}(\mathcal{E}_W) + 8\gamma^\top \mathcal{E}_W S + 8 \text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]. \end{aligned}$$

Consequently the asymptotic deficiency of IREG relative to REG, computed from the expansion MSE, is $\Delta = D_p/V^*$ with V^* as in (5).

The proof is in Appendix A with numerical verification and demonstration in Appendix B.

The expression for D_p combines four key blocks: a covariate shape block scaled by the linear HTE γ ; terms crossing γ with residual asymmetry; residual heteroskedasticity; and nonlinear misspecification. No single term determines the sign; the terms matter through their signs and magnitudes. For instance, when $\gamma = 0$ all terms involving γ vanish and

$$D_p = 8 \text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2].$$

While $\text{tr}(\mathcal{E}^2) = \|\mathcal{E}\|_F^2 \geq 0$, the term $S^\top \mathbb{E}[Z\varepsilon^2]$ can have either sign, so Δ can be positive or negative in this case.

Remark 3 (Bias and variance components). Two of the terms in D_p are a squared-bias difference; the fact that regression adjustment has a bias of order $1/n$ is classic [Freedman, 2008, Lin, 2013]. Let $\tilde{\tau}_n$ be the Taylor polynomial of Definition 1. Its expanded bias is $\mathbb{E}[\tilde{\tau}_n - \tau] = \frac{1}{2} \text{tr}(H\Sigma_n) + O(n^{-2})$, where H is the Hessian of the estimator's coefficient function and Σ_n the covariance of the per-arm means. Evaluating this leading term for the two estimators (Lemma 14 in Appendix A) gives

$$\mathbb{E}[\tilde{\tau}_{n,\text{REG}} - \tau] = -\frac{4 \text{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}), \quad \mathbb{E}[\tilde{\tau}_{n,\text{IREG}} - \tau] = -\frac{4 \text{tr}(\mathcal{E}_W)}{n} + O(n^{-2}),$$

so that $n^2\{\text{Bias}_{\text{IREG}}^2 - \text{Bias}_{\text{REG}}^2\} = -(\gamma^\top S)^2 - 8(\gamma^\top S) \text{tr}(\mathcal{E}_W)$ to leading order, exactly the $-Q - 8B_P$ part of D_p in the notation of Appendix A; the remaining terms are variance contributions.

To interpret the scale of Δ we define the variance share due to HTE as

$$R_\tau^2 = \frac{\|\gamma\|^2}{\|\gamma\|^2 + 4\mathbb{E}[\varepsilon_i^2]} = \frac{\|\gamma\|^2}{V^*}.$$

Consider the balanced doubly robust pseudo-outcome built from the linear working models $\mu_w(x) = \alpha_{(w)}^* + x^\top \beta_{(w)}^*$,

$$\phi_i = 2W_i\{Y_i - \mu_1(X_i)\} - 2(1 - W_i)\{Y_i - \mu_0(X_i)\} + \mu_1(X_i) - \mu_0(X_i).$$

Under the canonical representation $\phi_i - \tau = \gamma^\top Z_i + \xi_i$ with $\xi_i = 2W_i\varepsilon_i(1) - 2(1 - W_i)\varepsilon_i(0)$, we have $\mathbb{E}[Z_i\xi_i] = 0$, $\text{Var}[\xi_i] = 4\mathbb{E}[\varepsilon_i^2]$ and $\text{Var}[\gamma^\top Z_i] = \|\gamma\|^2$, so R_τ^2 is the population R^2 from linearly predicting the treatment-effect pseudo-outcome from Z_i . As noted in Section 3.2, R_τ^2 is also the fraction of the population-estimand sampling variance of $\hat{\tau}_{\text{IREG}}$ that the default interacted-regression standard error omits under superpopulation sampling.

Corollary 4. *In addition to Assumption 1, suppose the potential outcomes satisfy a well-specified linear model with homoskedastic errors, $\mathbb{E}[\varepsilon_i(w) \mid X_i] = 0$ and $\mathbb{E}[\varepsilon_i^2(w) \mid X_i] = \sigma_w^2$ for $w \in \{0, 1\}$. Then $\mathcal{E} = \mathcal{E}_W = 0$ and $\mathbb{E}[Z\varepsilon^2] = 0$. For $\gamma \neq 0$,*

$$\Delta = R_\tau^2 \left[-(2p+3) + \frac{\gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2} \right]. \quad (9)$$

In the Gaussian sub-case $K = (p+2)I_p$, $S = 0$ and $S_\gamma = 0$, so that $\Delta = -(p+1)R_\tau^2$. For $\gamma = 0$, the corresponding expression follows directly from Theorem 2.

In the Gaussian homoskedastic case, the second-order coefficient favors IREG and is linear in the number of covariates, with p fixed as $n \rightarrow \infty$. This complicates the parameter-count heuristic of Zhao and Ding [2023]. Importantly though, the effect is a second-order statement with fixed p , and the theorem does not cover the regimes in which p grows with n studied by Lei and Ding [2021], Chiang et al. [2025]. The following special case makes this mechanism more transparent.

Remark 5 (Independent symmetric coordinates). Suppose that, in addition to the conditions of Corollary 4, the whitened covariates have independent coordinates with symmetric marginals and γ loads on coordinate j only. Then $S = 0$ and $S_\gamma = 0$, while $\gamma^\top K \gamma = (\kappa_j + p - 1)\|\gamma\|^2$ with $\kappa_j = \mathbb{E}[Z_j^4]$, so

$$\Delta = R_\tau^2 (\kappa_j - p - 4).$$

When $R_\tau^2 > 0$, $\Delta > 0$ exactly when the kurtosis of the covariate driving the heterogeneity exceeds $p+4$; the Gaussian case $\kappa_j = 3$ recovers $\Delta = -(p+1)R_\tau^2$.

Corollary 6. *In addition to Assumption 1, suppose the potential outcomes follow*

$$Y_i(w) = \alpha_{(w)}^* + X_i^\top \beta_{(w)}^* + \eta_i \|Z_i\|,$$

where $\mathbb{E}[\eta_i] = 0$, $\mathbb{E}[\eta_i^2] = \nu^2$ and η_i is independent of (Z_i, W_i) . Then $\mathbb{E}[\varepsilon_i(w) \mid Z_i] = 0$ and $\mathbb{E}[\varepsilon_i^2(w) \mid Z_i] = \nu^2 \|Z_i\|^2$, where $\|\cdot\|$ is the Euclidean norm, so that $\mathbb{E}[Z_i \varepsilon_i^2(w)] = \nu^2 S$ and $\mathbb{E}[\varepsilon_i^2] = \nu^2 p$. For $\gamma \neq 0$,

$$\Delta = R_\tau^2 \left[-(2p+3) + \frac{\gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2} \right] + \frac{8\nu^2 S^\top S}{\|\gamma\|^2 + 4\nu^2 p}. \quad (10)$$

For $\gamma = 0$, the corresponding expression again follows directly from Theorem 2.

Interpretation and scaling. It is useful to consider how the terms in (10) compare in practice. The treatment-effect signal relative to residual variance is small in many applications, so terms multiplied by R_τ^2 rarely dominate unless the moment factors are large. The bracketed component is of order p in the leading cases we have examined—it equals $-(p+1)$ under Gaussian covariates and $\kappa_j - p - 4$ in the setting of Remark 5, which is negative unless the kurtosis exceeds $p+4$ —and its numerator grows with the higher moments of the covariate distribution, so it is sensitive to departures from Gaussianity in either direction.

The remaining term is nonnegative, since $8\nu^2 \|S\|^2 / (\|\gamma\|^2 + 4\nu^2 p) \geq 0$, and for $\nu > 0$ it vanishes if and only if $S = \mathbb{E}[Z\|Z\|^2] = 0$. Right-skewed covariates with a long upper tail, such as measures of user activity, engagement or spending, can produce a large third-moment vector S after centering and whitening, in which case this term can dominate the covariate-shape block. The scalar case makes this transparent: with $p = 1$, Corollary 6 gives

$$\Delta = R_\tau^2 \left\{ \mathbb{E}[Z_i^4] - 4\mathbb{E}[Z_i^3]^2 - 5 \right\} + \frac{8\nu^2 \mathbb{E}[Z_i^3]^2}{\gamma^2 + 4\nu^2}.$$

Using Pearson’s inequality $\mathbb{E}[Z_i^4] \geq \mathbb{E}[Z_i^3]^2 + 1$ [see, e.g., Rohatgi and Székely, 1989] and $8\nu^2/(\gamma^2 + 4\nu^2) = 2(1 - R_\tau^2)$,

$$\Delta \geq -4R_\tau^2 + (2 - 5R_\tau^2)\mathbb{E}[Z_i^3]^2.$$

So when $R_\tau^2 < 2/5$ the deficiency is positive as soon as $\mathbb{E}[Z_i^3]^2 > 4R_\tau^2/(2 - 5R_\tau^2)$, and the lower bound grows linearly in $\mathbb{E}[Z_i^3]^2$. The converse caveat, documented in Appendix C, is that the same third and fourth moments that make Δ large also slow the convergence of the expansion: in the lognormal family we study, the regime in which Δ is large is also the regime in which $1 + \Delta/n$ materially overstates the finite-sample MSE ratio at moderate n .

Estimating D_p . Theorem 2 suggests estimating D_p and choosing between REG and IREG accordingly. While promising, this rule cannot be taken at face value; it is precisely the situations where $\hat{D}_p$ would be noisiest and so least reliable where the difference would matter the most. We leave refinement of this idea to future work.

3 Feasible Inference

The above results concern point estimation. In practice both estimators are reported with confidence intervals, whose behavior depends on the variance estimator as much as on the estimator. For centered IREG under superpopulation sampling there are two distinct sources of difficulty to address: finite-sample variance estimation, which the arm-specific fit makes sensitive to skewness and leverage, and an estimand-level variance component that conventional fixed-design inference omits. This section separates them with three simulation experiments. We first remove the estimand-level component by setting $G = 0$ (Section 3.3), then vary treatment-effect heterogeneity in a clean design to isolate it (Section 3.4), and finally combine the two across a diverse selection of designs (Section 3.5). Throughout, the MSE ratio compares the point estimators, while coverage and interval width describe an estimator together with its variance estimator.

3.1 Implementation and variance estimators

We use the R package `estimatr` [Blair et al., 2024]. REG is `lm_robust(y ~ w + x)`, and IREG is `lm_lin(y ~ w, covariates = ~ x)`, which centers the covariates at their sample means, interacts them with treatment, and reports $\hat{\tau}_{\text{IREG}}$ as the coefficient on w . We compare two heteroskedasticity-robust standard errors. HC1, which scales the squared residuals by $n/(n-k)$ for a regression with k columns, is the conventional benchmark: it is Stata’s default and is used in 98.1% of 40,571 heteroskedasticity-robust Stata regressions in reproduction packages of leading economics journals [Kranz, 2024]. HC3 inflates each squared residual by $(1 - H_{ii})^{-2}$, approximates the jackknife [MacKinnon and White, 1985] and is recommended for small and moderate samples [Long and Ervin, 2000]; it is the leverage-sensitive benchmark. Intervals use t critical values with $n - k$ degrees of freedom. The contrast matters most for IREG: the interacted specification amounts to separate least-squares fits in the two arms, each on about $n/2$ units, so an extreme covariate value has substantially greater within-arm leverage than in the pooled REG fit [Chesher and Jewitt, 1987].

3.2 The superpopulation variance component

For the centered IREG implementation, the fixed-design variance treats $\bar{X}$ as fixed. Under the superpopulation sampling of Section 2, $\hat{\tau}_{\text{IREG}} = \hat{\tau} + \bar{X}^\top \hat{\delta}$, and the randomness of $\bar{X}$, together with heterogeneous treatment effects, contributes

$$\frac{G}{n}, \quad G = (\beta_{(1)}^* - \beta_{(0)}^*)^\top \Sigma_X (\beta_{(1)}^* - \beta_{(0)}^*) = \|\gamma\|^2 = R_\tau^2 V^*,$$

with $\Sigma_X = \text{Var}[X_i]$, to its variance for the population ATE. This is a known additional component arising from a population rather than a sample estimand [Imbens and Wooldridge, 2009, Negi and Wooldridge, 2021]. Omitting G/n gives IREG an asymptotic 95% coverage of $2\Phi(1.96\sqrt{1 - R_\tau^2}) - 1$. Zhao and Ding [2021] propose the plug-in correction

$$\hat{G} = \hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}, \quad \hat{V}_{\text{SP}} = \hat{V}_{\text{IREG}} + \frac{\hat{G}}{n}. \quad (11)$$

The estimator $\hat{G}$ is consistent for G , so $\hat{G}/n$ estimates the G/n component. It is biased upward in finite samples: under homoskedastic, well-specified errors with variance σ^2 , $\mathbb{E}[\hat{G}/n] \approx G/n + 4p\sigma^2/n^2$, so the plug-in is positive even when $G = 0$, and heteroskedasticity and heavy tails can make this bias quite large. We therefore report, beside the feasible correction, an oracle that adds the true G/n . Our simulations address how much the correction matters as heterogeneity grows, and how it interacts with the finite-sample effects of the regression specification.

3.3 Finite-sample effects when $G = 0$

In these designs, $\gamma = 0$ and $Y_i(1) - Y_i(0) = 1$, so $G = 0$ and the sample and population ATEs coincide. These designs therefore isolate the finite-sample behavior of the regression specification and the variance estimator. Treatment is assigned by a balanced CRT with $n = 500$, and

$$Y_i(0) = 0.9 X_i + \sigma(X_i) u_i, \quad Y_i(1) = Y_i(0) + 1, \quad \sigma(X) = 1 + X, \quad u_i \sim N(0, 1),$$

with a right-skewed covariate $X_i \sim \text{LogNormal}(0, s^2)$ such as prior engagement or spending; the introductory example is this design with $s = 1.1$, which is not tabulated. Each design uses 20,000 replications, so the Monte Carlo standard error of every coverage is at most 0.27 percentage points. The simulations call `estimatr`'s own fitting routine with the design matrices that `lm_robust` and `lm_lin` build (Appendix D).

Table 1 shows that IREG+HC1 coverage deteriorates steadily as the covariate becomes more skewed, from 94.5% at $s = 0.6$ to 89.9% at $s = 1.5$ and 83.0% at $s = 2$. HC3 substantially improves IREG's coverage, to 94.7% at $s = 1.5$, but it too falls below nominal in the heaviest designs (93.9% at $s = 2$). REG is much more stable: at the 95% level REG+HC1 stays between 94.7% and 95.4% in every $\gamma = 0$ design we ran (Appendix F), although in the most skewed design it undercovers at 80% and overcovers at 99%, so its near-nominal 95% coverage there is partly a crossing point.

The widths and the MSE ratio show that IREG's problem is not the size of its standard error. The median HC1 widths of IREG and REG are nearly equal throughout (6.87 and 6.80 at $s = 2$), yet IREG's MSE exceeds REG's by a factor of 2.00 at $s = 2$: the issue lies in the point estimator itself. The pattern is consistent with the leverage mechanism of Chesher and Jewitt [1987]. IREG fits each arm separately, so a rare extreme covariate value has roughly

Table 1: Root-mean-squared error of the point estimators, and median width and coverage (%) of nominal 95% intervals, when $\gamma = 0$ (so $G = 0$): $X \sim \text{LogNormal}(0, s^2)$, $\sigma(X) = 1 + X$, $p = 1$, $n = 500$, 20,000 replications.

		IREG (lm_lin)						REG (lm_robust)					
		RMSE		HC1		HC3		RMSE		HC1		HC3	
s	skew Z			width	cov.	width	cov.			width	cov.	width	cov.
0.6	2.3	0.211	0.82	94.5	0.82	94.8	0.210	0.81	94.7	0.82	94.9		
0.8	3.7	0.242	0.93	94.4	0.95	95.1	0.240	0.93	94.7	0.94	95.1		
1.0	6.2	0.301	1.13	94.2	1.18	95.4	0.295	1.13	95.1	1.15	95.8		
1.2	11.2	0.420	1.47	92.6	1.58	95.1	0.401	1.46	94.8	1.51	95.8		
1.5	33.5	0.812	2.43	89.9	2.81	94.7	0.721	2.41	95.0	2.57	96.7		
1.8	136.4	1.864	4.43	86.1	5.56	94.1	1.464	4.39	95.1	4.81	97.4		
2.0	414.4	3.482	6.87	83.0	9.14	93.9	2.465	6.80	95.3	7.57	97.9		

twice the leverage it would have in REG’s pooled fit. HC1 does not undo this shrinkage; the factor $(1 - H_{ii})^{-2}$ in HC3 largely does.

Taking this idea to the extreme, an arm containing a single extreme covariate value fits it exactly: the observation has leverage one, its residual is zero, and no residual-based variance estimator can recover its contribution to the variance of $\hat{\tau}_{\text{IREG}}$. Appendix F analyzes this case.

3.4 Treatment-effect heterogeneity and the superpopulation correction

To isolate G/n we vary treatment-effect heterogeneity in a design with no leverage problems: $p = 5$, $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$, $Y_i(0) = 0.9 \sum_j X_{ij} + u_i$ and $Y_i(1) = Y_i(0) + 1 + \gamma^* X_{i1}$, with $u_i \sim N(0, 1)$ shared across arms, $n = 500$ and 20,000 replications. Only the slope γ^* varies. Since $\Sigma_{11} = 1$ and $\mathbb{E}[\varepsilon^2] = 1$, $G = (\gamma^*)^2$ and $R_\tau^2 = (\gamma^*)^2 / \{(\gamma^*)^2 + 4\}$; we choose γ^* so that R_τ^2 runs from 0 to 0.9, and compute R_τ^2 from the generated design.

Table 2: Coverage (%) of nominal 95% intervals for the population ATE and median interval width as treatment-effect heterogeneity grows, in a Gaussian, homoskedastic, well-specified design ($p = 5$, $n = 500$). “Plug-in” adds $\hat{G}/n$ as in (11); “oracle” adds the true G/n ; “asymptotic” is $2\Phi(1.96\sqrt{1 - R_\tau^2}) - 1$, the limit of uncorrected IREG coverage.

R_τ^2	Coverage					Median width		
	IREG, HC1				REG	IREG, HC1		REG
	none	asymptotic	plug-in	oracle	HC1	none	plug-in	HC1
0.00	94.9	95.0	95.0	94.9	94.9	0.353	0.355	0.353
0.10	93.7	93.7	95.1	95.0	94.9	0.353	0.374	0.372
0.25	91.1	91.0	95.0	94.9	94.9	0.353	0.408	0.407
0.50	83.3	83.4	95.1	95.1	95.1	0.353	0.499	0.499
0.70	71.8	71.7	95.2	95.2	95.2	0.353	0.643	0.645
0.85	55.4	55.2	94.9	95.0	95.0	0.353	0.908	0.912
0.90	47.1	46.5	95.1	95.2	95.1	0.353	1.112	1.117

Table 2 and Figure 1 show the pattern. With $R_\tau^2 = 0$ every interval is close to nominal and the oracle does nothing. As R_τ^2 grows, uncorrected IREG+HC1 undercovers more and more, from 93.7% at $R_\tau^2 = 0.1$ to 83.3% at 0.5 and 47.1% at 0.9, tracking its asymptotic

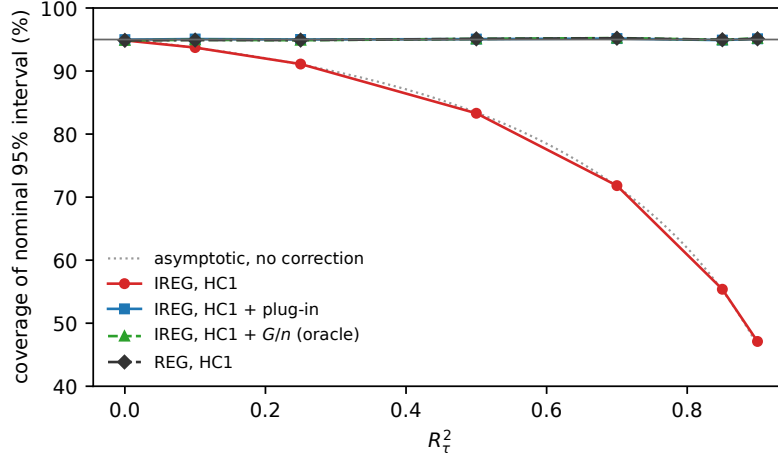


Figure 1: Coverage of nominal 95% intervals against R^2_τ in the design of Table 2. The dotted curve is the asymptotic coverage of IREG without correction.

coverage. The oracle correction restores nominal coverage at every level (94.9–95.2%), and the feasible correction stays within 0.1 points of the oracle, so in this design the plug-in estimates the population component it targets. REG+HC1, whose robust variance does not omit G/n , covers between 94.9% and 95.2% throughout. The corrected IREG interval widens as the component grows: at $R^2_\tau = 0.9$ its median width is 1.11 against 0.35 without the correction and 1.12 for REG+HC1; the extra width is the genuine population uncertainty. HC3 does not address the omission: at $R^2_\tau = 0.9$ IREG+HC3 covers 47.6%, barely more than HC1. HC3 corrects leverage-driven finite-sample distortion of the variance estimate, whereas the G/n correction accounts for sampling uncertainty that the fixed-design variance does not describe; they address different problems. In this well-specified Gaussian design, finite-sample coverage differences at the 95% level are small, and the plug-in tracks the population component it targets.

3.5 Combined effects across designs

Table 3 compares all four procedures across 14 designs in which the two mechanisms can coexist. IREG is paired with the plug-in correction. The selection emphasizes small heterogeneity and ranges from Gaussian designs to skewed, heavy-tailed and rare-value covariates, with MSE ratios from 0.97 to 1.95. The scalar lognormal designs are those of Section 3.3 with a treatment-effect slope added; the others reuse the designs of the correction study (Appendix F). All have $n = 500$ and 20,000 replications.

With little or no heterogeneity, every procedure covers at least 93% at the 95% level: IREG+HC1 with the plug-in covers between 93.7% and 95.0%, and REG+HC1 between 93.2% and 95.0%. The plug-in’s upward bias happens to offset the leverage-driven understatement of HC1 documented in Section 3.3, and the resulting coverage is bought with width. The IREG+HC1 interval with the plug-in is up to 1.14 times as wide as REG+HC1’s at the median, and IREG+HC3 with the plug-in, which overcovers (up to 97.8%), is up to 1.30 times as wide as REG+HC3’s. REG reaches the same coverage with shorter intervals, consistent with its smaller MSE in these designs. In the Gaussian design the plug-in’s bias is small (1.4 times G/n) and the two estimators’ intervals have the same width.

Table 3: Root-mean-squared error of the point estimators, and median width and coverage (%) of nominal 95% intervals for the population ATE, for IREG with the plug-in correction and for REG, each with HC1 and HC3 ($n = 500$; “het.” denotes heteroskedastic noise). Designs are defined in Appendix F.

Design	R^2_τ	IREG + plug-in						REG					
		RMSE	HC1		HC3			RMSE	HC1		HC3		
			width	cov.	width	cov.			width	cov.	width	cov.	
<i>Small or no heterogeneity</i>													
Gaussian	0.02	0.091	0.36	95.0	0.36	95.2	0.091	0.36	94.8	0.36	95.0		
Misspecified, $G = 0$	0	0.098	0.38	94.9	0.39	95.3	0.098	0.38	94.8	0.38	95.0		
Skewed, het., $G = 0$	0	0.167	0.65	95.0	0.67	95.9	0.165	0.64	95.0	0.65	95.5		
Heavy-tailed, het.	0.02	0.175	0.64	94.7	0.68	96.2	0.168	0.63	94.7	0.64	95.5		
Lognormal $s = 1.2$	0.01	0.425	1.52	94.6	1.63	96.1	0.408	1.47	94.6	1.53	95.7		
Lognormal $s = 1.5$	0.002	0.798	2.55	94.7	2.91	97.0	0.715	2.40	94.9	2.55	96.7		
Lognormal $s = 1.5$	0.05	0.839	2.62	93.7	2.98	96.2	0.753	2.46	93.2	2.61	95.1		
Lognormal $s = 1.8$	0.009	2.826	7.26	94.3	8.85	97.3	2.196	6.57	94.3	7.19	96.6		
Lognormal $s = 2$	0.002	3.433	7.76	94.8	9.84	97.8	2.458	6.80	95.0	7.56	97.6		
<i>Moderate heterogeneity</i>													
Lognormal $s = 1.5$	0.17	0.901	2.81	92.0	3.16	94.9	0.826	2.64	90.2	2.81	92.0		
Rare extreme values	0.20	0.100	0.40	94.0	0.40	94.2	0.102	0.40	94.7	0.41	95.5		
<i>Large heterogeneity</i>													
Heavy-tailed, het.	0.37	0.220	0.75	94.2	0.79	95.2	0.212	0.73	91.9	0.76	93.1		
Lognormal $s = 1.5$	0.65	1.451	4.46	89.8	4.75	91.4	1.380	4.17	84.1	4.52	86.0		
Gaussian	0.69	0.161	0.64	95.1	0.64	95.2	0.162	0.64	95.0	0.64	95.3		

Table 4: Coverage (%) at nominal 80%, 95% and 99% in the designs of Table 3 with small but nonzero heterogeneity: IREG with HC1 plus the true G/n (oracle) or plus the plug-in, and REG with HC1. “Plug-in/ (G/n) ” is the ratio of the average plug-in to the true component; the last column is the median width of the IREG plug-in interval relative to REG+HC1’s.

Design	R^2_τ	Plug-in $/(G/n)$	IREG, HC1 + oracle			IREG, HC1 + plug-in			REG, HC1			Width ratio
			80	95	99	80	95	99	80	95	99	
Gaussian	0.02	1.4	79.9	94.9	99.0	80.1	95.0	99.1	79.9	94.8	99.0	1.00
Heavy-tailed, het.	0.02	5.7	77.5	93.5	98.4	79.1	94.7	98.9	79.0	94.7	98.9	1.02
Lognormal $s = 1.2$	0.01	6.7	76.3	92.9	98.1	78.4	94.6	98.9	78.6	94.6	98.9	1.03
Lognormal $s = 1.5$	0.002	71.0	72.6	90.4	96.8	77.0	94.7	99.3	77.8	94.9	99.3	1.06
Lognormal $s = 1.5$	0.05	4.1	73.2	90.5	96.8	76.6	93.7	98.5	76.4	93.2	98.3	1.06
Lognormal $s = 1.8$	0.009	25.6	68.3	86.4	94.2	75.5	94.3	99.1	76.4	94.3	99.0	1.11
Lognormal $s = 2$	0.002	90.2	65.3	83.5	92.1	75.1	94.8	99.3	76.1	95.0	99.3	1.14

Table 4 shows why the plug-in is not a general correction for IREG+HC1. With small R_τ^2 the plug-in interval is miscalibrated across nominal levels. In the $s = 2$ design, for example, its coverage is 75.1%, 94.8%, and 99.3% at nominal 80%, 95%, and 99%, respectively; the oracle gives 65.3%, 83.5%, and 92.1%. The plug-in’s 95% coverage therefore reflects finite-sample variance inflation rather than estimation of a substantial population component.

The moderate- and large-heterogeneity designs are included for illustration, and there the ordering reverses. In the skewed designs with substantial heterogeneity, REG is the procedure that undercovers: in the lognormal $s = 1.5$ design at $R_\tau^2 = 0.65$ it covers 84.1% with HC1 and 86.0% with HC3, against 89.8% and 91.4% for IREG with the plug-in, and the heavy-tailed design at $R_\tau^2 = 0.37$ shows the same ordering (91.9% and 93.1% against 94.2% and 95.2%). REG’s robust variance retains the heterogeneity term, but retaining it is not the same as estimating it well: when a large treatment-effect slope is combined with a skewed covariate, the pooled fit attributes the arm-specific slope difference to its residuals, and its variance estimate degrades faster than IREG’s corrected one. In these designs the plug-in and HC3 address different components of the coverage problem: the plug-in accounts for population-ATE uncertainty, while HC3 addresses leverage-driven finite-sample variance estimation.

4 Discussion

Our results distinguish three sources of differences between simple and interacted regression adjustment: the higher-order behavior of the point estimator, uncertainty about the population estimand, and finite-sample behavior of the variance estimator. Together they give a more specific practical picture than first-order efficiency alone.

Point estimation. Under balanced assignment, REG and IREG have the same first-order asymptotic variance, so the comparison between them is settled at second order, and there we find no universal ordering. Theorem 2 identifies the features that determine the next-order comparison: treatment-effect heterogeneity, covariate shape, residual heteroskedasticity, and nonlinear misspecification. In the well-specified Gaussian case, the second-order coefficient favors IREG and, with p fixed as $n \rightarrow \infty$, is linear in p . More generally, however, skewness and heavy tails can reverse this comparison, particularly when they interact with heteroskedasticity or misspecification. In our scalar lognormal designs, this produces deficiencies of 12, 41, and 166 observations at $s = 0.8, 1, 1.2$, respectively, and at $n = 500$ the simulated MSE of IREG exceeds that of REG by 4.3% and 9.2% in the latter two designs. The most heavy-tailed designs show still larger finite-sample gaps, although there the n^{-2} expansion no longer accurately predicts their precise magnitude. The direction of the comparison nevertheless agrees with the expansion in every design we examine, showing that D_p can remain informative about which estimator has lower MSE even when the approximation is no longer quantitatively accurate.

Inference when treatment-effect heterogeneity is small. The practical picture is clearer when R_τ^2 is small. Here the superpopulation component G/n is small, while finite-sample leverage can still be substantial. In our skewed and heteroskedastic designs, IREG with HC1 can substantially undercover because the arm-specific fits concentrate leverage on extreme covariate values. HC3 substantially improves coverage for IREG in most such designs. The feasible plug-in can also bring coverage close to nominal, but when G is small its upward finite-sample bias is large relative to the component it is intended to estimate. REG with HC3 is close to nominal in these designs, while IREG with the plug-in and HC3 overcovers (up to 97.8%); REG also has lower MSE and, in the skewed designs, shorter intervals.

Inference when treatment-effect heterogeneity is large. The interpretation changes once linear treatment-effect heterogeneity is substantial. For the population ATE, centered IREG has an additional first-order variance component $G/n = R_\tau^2 V^*/n$ that its conventional robust

variance omits. The feasible plug-in tracks this component well in our Gaussian calibration design, and adding it restores nominal coverage at the cost of wider intervals that reflect genuine population uncertainty. HC3 alone does not address this omission. In designs where both heterogeneity and leverage are substantial, the two issues coexist, and both procedures can undercover in the most extreme designs; there it is REG that undercovers more, as Section 3.5 reports. HC3 improves coverage for both, while the plug-in addresses the separate population-ATE component.

A practical distinction between REG and IREG. These results suggest a simple way to interpret the choice between the two procedures. IREG has no universal second-order disadvantage and can be better in well-specified settings, particularly with Gaussian covariates. But in the small-heterogeneity, skewed or heavy-tailed settings represented in our simulations, REG combines lower point-estimation MSE with robust inference after HC3, whereas IREG only achieves nominal coverage at the cost of noticeably wider confidence intervals. REG also has the practical advantage that its regression adjustment does not require a separate variance component when moving between the fixed-population and superpopulation interpretations considered here.

Limitations of our work. These conclusions are conditional on the regimes and designs studied here, rather than a universal ranking of REG and IREG. The theorem holds with fixed p and does not cover the many-covariate regimes studied by Lei and Ding [2021] and Chiang et al. [2025]. The second-order result concerns the expansion MSE of Definition 1; connecting it to exact or trimmed moments requires additional conditions that we do not verify. The inference results are numerical, and we do not derive higher-order expansions for HC coverage.

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A Proofs

Throughout this appendix, indices i, j, k, l, a, b, c, d for $Z, T_3, T_4, \mathcal{E}$ and P refer to coordinates of the whitened covariate vector. We write $G = \|\gamma\|^2$.

A.1 Second-order expansion

Under complete randomization with $n_1 = m$ treated units, write $U_i^{(w)}$ for the vector containing $X_i, Y_i(w)$ and the distinct entries of $X_i X_i^\top$ and $X_i Y_i(w)$, and let $\bar{U}_1$ and $\bar{U}_0$ be the averages of $U_i^{(1)}$ over treated units and of $U_i^{(0)}$ over control units, with population means μ_1 and μ_0 .

Lemma 7 (Two-sample representation). *Let $(X_i, Y_i(0), Y_i(1))$ be IID and let the assignment vector W be independent of them, with $\sum_i W_i = m$. Conditionally on W , the treated observations $\{(X_i, Y_i(1)) : W_i = 1\}$ and the control observations $\{(X_i, Y_i(0)) : W_i = 0\}$ are independent IID samples of sizes m and $n - m$ from the laws of $(X, Y(1))$ and $(X, Y(0))$ respectively. Both $\hat{\tau}_{\text{REG}}$ and $\hat{\tau}_{\text{IREG}}$ are of the form $h_{j,r}(\bar{U}_1, \bar{U}_0)$ with $r = m/n$, where $h_{j,r}$ is infinitely differentiable in a neighborhood of (μ_1, μ_0) and its derivatives at (μ_1, μ_0) are rational functions of r without poles in $(0, 1)$.*

Proof. Given W , the units remain IID because W is independent of them; the treated and control index sets are fixed and disjoint, so the two blocks are independent IID samples, observed through $Y_i(1)$ and $Y_i(0)$ respectively. Both estimators are rational functions of per-arm means, second moments and cross-moments: $\hat{\tau}_{\text{IREG}}$ is the difference of the arm-wise intercepts after centering at $\bar{X} = r\bar{X}_1 + (1-r)\bar{X}_0$, with arm-wise least-squares slopes $\mathcal{S}_w^{-1} s_w$ built from $\bar{U}_w$; $\hat{\tau}_{\text{REG}}$

uses the pooled within-arm slope, whose Gram matrix is $r\mathcal{S}_1 + (1-r)\mathcal{S}_0$. The Gram matrices are invertible at (μ_1, μ_0) because $\Sigma \succ 0$, which gives the smoothness and the dependence on r . $\square$

Lemma 8 (Conditional expansion). *Under Assumption 1, let $\text{MSE}_{\text{CRT},r}^{\text{exp}}(\hat{\tau}_j)$ be the second moment of the third-order Taylor polynomial of $h_{j,r}$ at (μ_1, μ_0) . For every $\eta \in (0, \frac{1}{2})$,*

$$\text{MSE}_{\text{CRT},r}^{\text{exp}}(\hat{\tau}_j) = \frac{V_j(r)}{n} + \frac{c_j(r)}{n^2} + O(n^{-3}), \quad j \in \{\text{REG}, \text{IREG}\}, \quad (12)$$

uniformly in $r = m/n \in [\frac{1}{2} - \eta, \frac{1}{2} + \eta]$, with V_j and c_j infinitely differentiable in r .

Proof. The Taylor polynomial is a polynomial of degree three in $(\bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$ whose coefficients are smooth in r (Lemma 7); its square has degree six. By Lemma 7 the two blocks are independent IID samples, so each mixed moment of the centered per-arm means is a finite sum of products of joint cumulants of $U^{(1)}$ and $U^{(0)}$ divided by powers of $m = rn$ and $(n - m) = (1 - r)n$. Moments up to order six are finite because the entries of $U^{(w)}$ are products of at most two coordinates of $(X_i, Y_i(w))$ and Assumption 1 bounds their twelfth moments. Collecting terms gives a finite expansion in integer powers of $1/n$ whose coefficients are rational in r without poles on $[\frac{1}{2} - \eta, \frac{1}{2} + \eta]$; the terms beyond n^{-2} are $O(n^{-3})$ uniformly on that interval. $\square$

Lemma 9 (Bernoulli and conditional expansions). *Under Assumption 1 with $W_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2)$ independent of the units, let $\bar{W} = n^{-1} \sum_i W_i$. For every $\eta \in (0, \frac{1}{2})$,*

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_j) = \mathbb{E} \left[\text{MSE}_{\text{CRT},\bar{W}}^{\text{exp}}(\hat{\tau}_j) \mathbf{1}\{|\bar{W} - \frac{1}{2}| \leq \eta\} \right] + O(n^{-5/2}).$$

Proof. Let $\mathcal{P}_{\text{Bern}}$ be the Taylor polynomial of Definition 1 under Bernoulli assignment and $\mathcal{P}_{\bar{W}}$ the conditional polynomial of Lemma 8 evaluated at $r = \bar{W}$. Every coordinate of $\bar{\mathcal{V}} - \mu$ is a polynomial of degree at most two in $(\bar{W} - \frac{1}{2}, \bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$ and is linear in the per-arm means; for instance,

$$n^{-1} \sum_i W_i U_i^{(1)} - \frac{1}{2} \mu_1 = (\bar{W} - \frac{1}{2}) \mu_1 + \frac{1}{2} (\bar{U}_1 - \mu_1) + (\bar{W} - \frac{1}{2}) (\bar{U}_1 - \mu_1).$$

Fix $\eta \in (0, \frac{1}{2})$ and consider the event $|\bar{W} - \frac{1}{2}| \leq \eta$. On this event, write

$$\hat{\tau}_j = F_j(\bar{W} - \frac{1}{2}, \bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0),$$

where F_j is smooth. Both $\mathcal{P}_{\text{Bern}}$ and $\mathcal{P}_{\bar{W}}$ agree with the Taylor expansion of F_j at the origin through total degree three: $\mathcal{P}_{\text{Bern}}$ because the map from $(\bar{W}, \bar{U}_1, \bar{U}_0)$ to $\bar{\mathcal{V}}$ is polynomial, and $\mathcal{P}_{\bar{W}}$ because it is the conditional expansion with $r = \bar{W}$. Hence their difference consists only of terms of total degree at least four in quantities that are $O_{L^k}(n^{-1/2})$. Under Assumption 1,

$$\|\mathcal{P}_{\text{Bern}} - \mathcal{P}_{\bar{W}}\|_{L^2} = O(n^{-2}).$$

Also,

$$\|\mathcal{P}_{\text{Bern}} + \mathcal{P}_{\bar{W}} - 2\tau\|_{L^2} = O(n^{-1/2}),$$

so Cauchy–Schwarz gives

$$|\mathbb{E} [(\mathcal{P}_{\text{Bern}} - \tau)^2 - (\mathcal{P}_{\bar{W}} - \tau)^2] \mathbf{1}\{|\bar{W} - \frac{1}{2}| \leq \eta\}| = O(n^{-5/2}).$$

Finally, Hoeffding's inequality gives

$$\Pr(|\bar{W} - \tfrac{1}{2}| > \eta) \leq 2e^{-2n\eta^2},$$

so the complement contributes an exponentially small term, which is $O(n^{-5/2})$. This proves the claim. $\square$

The proof of Theorem 2 combines these lemmas, a reduction from the CRT to the Bernoulli design, and the Bernoulli calculation.

Theorem 10. *Suppose the Bernoulli design has second-order difference*

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}}}{n^2} + O(n^{-5/2}) \quad (13)$$

at $\pi = 1/2$, and let Assumption 1 hold. Then, under complete randomization with $n_1 = \lfloor n/2 \rfloor$ treated units,

$$\text{MSE}_{\text{CRT}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{CRT}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}} + 4G}{n^2} + O(n^{-3}). \quad (14)$$

Proof. Let $n_1 = \sum_{i=1}^n W_i$ under the Bernoulli design and write $r = n_1/n$. Conditional on $n_1 = m$, the treatment assignment has the same distribution as a CRT with treatment fraction r . Fix $\eta \in (0, \frac{1}{2})$. By Lemma 9, $\text{MSE}_{\text{Bern}}^{\text{exp}}$ is, up to $O(n^{-5/2})$, the average over n_1 of $\text{MSE}_{\text{CRT},r}^{\text{exp}}$ restricted to $|r - \frac{1}{2}| \leq \eta$, and by Lemma 8 the latter admits the expansion (12) uniformly on that set, with V_j and c_j infinitely differentiable. At treatment fraction r the first-order variance difference is, by Proposition 1,

$$V_{\text{IREG}}(r) - V_{\text{REG}}(r) = -\frac{(1-2r)^2}{r(1-r)}G,$$

and

$$\frac{(1-2r)^2}{r(1-r)} = 16u^2 + O(u^4), \quad u = r - \tfrac{1}{2}.$$

Under Bernoulli randomization with $\pi = 1/2$, $\mathbb{E}[u^2] = 1/(4n)$ and $\mathbb{E}[u^4] = O(n^{-2})$, so

$$\frac{1}{n}\mathbb{E}[V_{\text{IREG}}(r) - V_{\text{REG}}(r)] = -\frac{4G}{n^2} + O(n^{-3}).$$

For the second-order coefficients, a second-order Taylor expansion of $c_{\text{IREG}} - c_{\text{REG}}$ around $1/2$ together with $\mathbb{E}[u] = 0$ and $\mathbb{E}[u^2] = O(n^{-1})$ gives

$$\mathbb{E}[c_{\text{IREG}}(r) - c_{\text{REG}}(r)] = c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) + O(n^{-1}).$$

Averaging (12) over n_1 therefore yields

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) - 4G}{n^2} + O(n^{-5/2}),$$

so

$$D_{\text{Bern}} = c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) - 4G.$$

Under balanced complete randomization, $r = 1/2$ for even n , giving

$$D_{\text{CRT}} = c_{\text{IREG}}(1/2) - c_{\text{REG}}(1/2) = D_{\text{Bern}} + 4G.$$

For even n , Lemma 8 therefore gives directly

$$\text{MSE}_{\text{CRT}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{CRT}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}} + 4G}{n^2} + O(n^{-3}).$$

For odd n , $r - 1/2 = O(n^{-1})$. The first-order variance difference is quadratic in $r - 1/2$, so its contribution to the MSE is $O(n^{-3})$; the smoothness of $c_{\text{IREG}} - c_{\text{REG}}$ gives an $O(n^{-1})$ change in the second-order coefficient, which is also $O(n^{-3})$ after division by n^2 . Thus the same expansion holds for odd n . $\square$

Theorem 11. *Let Assumption 1 hold, except that $W_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2)$ independently of $(X_i, Y_i(0), Y_i(1))$. Define*

$$T_{3,ijk} = \mathbb{E}[Z_i Z_j Z_k], \quad T_{4,ijkl} = \mathbb{E}[Z_i Z_j Z_k Z_l], \quad q_i = T_{3,ijk} \delta_{jk}, \quad L_{ij} = T_{3,iab} T_{3,jab}, \quad (15)$$

write $S = q = \mathbb{E}[Z\|Z\|^2]$, and set $\mathcal{E} = \mathbb{E}[ZZ^\top \varepsilon]$ and $P = \mathbb{E}[ZZ^\top (W - \frac{1}{2})\varepsilon]$. For the scalar contractions

$$G = \|\gamma\|^2, \quad F = \gamma_i \gamma_j T_{4,ijkk}, \quad Q = (\gamma^\top q)^2, \quad \mathcal{R} = \gamma^\top L \gamma, \quad (16)$$

$$A_P = \gamma_i T_{3,ijk} P_{jk}, \quad B_P = (\gamma^\top q) \text{tr}(P), \quad C_P = \gamma^\top P q, \quad (17)$$

we have

$$\text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}_{\text{Bern}}^{\text{exp}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}}}{n^2} + O(n^{-5/2}), \quad (18)$$

where

$$D_{\text{Bern}} = -(2p + 7)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]. \quad (19)$$

Proof. Both REG and IREG contain the regressors 1, W and X . Subtracting any linear combination of these regressors from the outcome changes the fitted coefficient on W by the corresponding amount while leaving the estimation error unchanged, so we may subtract the population main effects and the term τW_i and work with the exact representation

$$Y_i = W_i X_i^\top \{\beta_{(1)}^* - \beta_{(0)}^*\} + \varepsilon_i.$$

Whitening as in (3) gives the canonical representation $Y_i = W_i \gamma^\top Z_i + \varepsilon_i$, and we write

$$W_i = \frac{1}{2} + \omega_i, \quad \omega_i \in \{-\frac{1}{2}, \frac{1}{2}\}, \quad \mathbb{E}[\omega_i] = 0, \quad \omega_i^2 = \frac{1}{4}.$$

Let $\mathcal{V}_i$ denote the vector of raw-moment coordinates needed to evaluate the REG or IREG coefficient map, and adopt the notation of Definition 1, so that $\hat{\tau} = g(\bar{\mathcal{V}})$ and $\tilde{\tau}_n$ is given by (8). Let

$$\Sigma_{ij}^{\mathcal{V}} = \text{Cov}(\mathcal{V}_i, \mathcal{V}_j), \quad \mu_{3,ijk}^{\mathcal{V}} = \mathbb{E}[(\mathcal{V}_i - \mu_i)(\mathcal{V}_j - \mu_j)(\mathcal{V}_k - \mu_k)]$$

be the covariance and third central-moment tensors of the moment vector. Expanding the square of (8) and using the second- and third-order moments of sample means gives the order- n^{-1} correction to $n \text{MSE}^{\text{exp}}$ as $\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3$ with

$$\mathcal{T}_1 = a_i H_{jk} \mu_{3,ijk}^{\mathcal{V}}, \quad (20)$$

$$\mathcal{T}_2 = \frac{1}{4} H_{ij} H_{kl} (\Sigma_{ik}^{\mathcal{V}} \Sigma_{jl}^{\mathcal{V}} + \Sigma_{il}^{\mathcal{V}} \Sigma_{jk}^{\mathcal{V}} + \Sigma_{ij}^{\mathcal{V}} \Sigma_{kl}^{\mathcal{V}}), \quad (21)$$

$$\mathcal{T}_3 = \frac{1}{3} a_i A_{3,jkl} (\Sigma_{ij}^{\mathcal{V}} \Sigma_{kl}^{\mathcal{V}} + \Sigma_{ik}^{\mathcal{V}} \Sigma_{jl}^{\mathcal{V}} + \Sigma_{il}^{\mathcal{V}} \Sigma_{jk}^{\mathcal{V}}), \quad (22)$$

the remaining terms being $O(n^{-3/2})$ in $n \text{MSE}^{\text{exp}}$ under Assumption 1. The difference between the order- n^{-2} MSE coefficients of REG and IREG is obtained by evaluating these three contractions for the two estimators.

The population Gram matrices and their inverses in the whitened coordinates are

$$A_R = \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & I_p & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}, \quad A_R^{-1} = \begin{pmatrix} 2 & 0 & -2 \\ 0 & I_p & 0 \\ -2 & 0 & 4 \end{pmatrix}, \quad A_I = \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & I_p & 0 & I_p/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & I_p/2 & 0 & I_p/2 \end{pmatrix},$$

$$A_I^{-1} = \begin{pmatrix} 2 & 0 & -2 & 0 \\ 0 & 2I_p & 0 & -2I_p \\ -2 & 0 & 4 & 0 \\ 0 & -2I_p & 0 & 4I_p \end{pmatrix}, \quad \theta_R = (0, \gamma/2, 0), \quad \theta_I = (0, 0, 0, \gamma).$$

Writing $\Omega = A^{-1}$ and $\theta = \Omega b$, differentiation of $A\theta = b$ gives the exact derivative identities

$$D\theta[h] = \Omega(db - dA\theta),$$

$$D^2\theta[h_1, h_2] = -\Omega dA_1\Omega(db_2 - dA_2\theta) - \Omega dA_2\Omega(db_1 - dA_1\theta),$$

$$D^3\theta[h_1, h_2, h_3] = \sum_{\varsigma \in \mathfrak{S}_3} \Omega dA_{\varsigma_1}\Omega dA_{\varsigma_2}\Omega(db_{\varsigma_3} - dA_{\varsigma_3}\theta).$$

REG's estimator is the treatment coefficient, $g_{\text{REG}}(v) = \theta_W(v)$. IREG's is $g_{\text{IREG}}(v) = \theta_W(v) + \bar{z}(v)^\top \theta_{WZ}(v)$, where $\bar{z}(v)$ is the full-sample covariate mean, a linear coordinate of v that vanishes at μ in whitened coordinates, and $\theta_{WZ} = \gamma$ at μ . The product rule gives

$$a_{\text{IREG}}[h] = D\theta_W[h] + D\bar{z}[h]^\top \gamma,$$

$$H_{\text{IREG}}[h_1, h_2] = D^2\theta_W[h_1, h_2] + D\bar{z}[h_1]^\top D\theta_{WZ}[h_2] + D\bar{z}[h_2]^\top D\theta_{WZ}[h_1],$$

$$A_{3,\text{IREG}}[h_1, h_2, h_3] = D^3\theta_W[h_1, h_2, h_3] + \sum_{\{i,j,k\}=\{1,2,3\}} D\bar{z}[h_i]^\top D^2\theta_{WZ}[h_j, h_k],$$

the sum running over the three choices of i . The covariance and third-moment blocks of $\mathcal{V}_i$ are products of moments of $(Z_i, \omega_i, \varepsilon_i)$ under the canonical representation of Section 2, with W_i marginally Bernoulli(1/2).

Substituting these derivative tensors and the covariance and third-moment blocks of $\mathcal{V}_i$ into (20)–(22), we reduce the resulting indexed expressions using tensor symmetries, dummy-index relabeling, and the identities implied by $\mathbb{E}[ZZ^\top] = I_p$ and $\omega_i^2 = 1/4$. This gives

$$\Delta\mathcal{T}_1 = 2G + 2F + 8\Xi_\gamma, \tag{23}$$

$$\Delta\mathcal{T}_2 = -3G - F - Q - \mathcal{R} - 8\Xi_\gamma - 8A_P - 8B_P, \tag{24}$$

$$\Delta\mathcal{T}_3 = -(2p + 6)G - 2\mathcal{R} - 8A_P + 8C_P + 8\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2], \tag{25}$$

where

$$\Xi_\gamma = \mathbb{E}[(\gamma^\top Z)\|Z\|^2\omega\varepsilon]$$

with $\omega = W - \frac{1}{2}$. The accompanying verification script enumerates the full Cartesian products of the expansion directions and reduces the resulting indexed expressions using only tensor symmetries, dummy-index relabeling, and the identities implied by $\mathbb{E}[ZZ^\top] = I_p$ and $\omega_i^2 = 1/4$, with the dimension p kept symbolic. The symbolic remainder in each of the three rows is identically

zero; the verification is implemented in `verification/general_p_symbolic_certificate.py` in the replication repository and runs in under a minute.

Summing the three rows, the Ξ_γ terms cancel and (19) follows.

The division of labour is as follows. Derived by hand and displayed above: the expansion terms (20)–(22), the Gram matrices and coefficients at μ , the derivative identities, the product-rule structure of the IREG centering functional, and the reduction rules. Verified symbolically by the accompanying implementation: the substitution of these objects into (20)–(22) and the reduction of the resulting indexed sums to the rows (23)–(25), with symbolic p and zero remainder. The resulting constant is checked independently by Monte Carlo in Appendix B. $\square$

Theorems 10 and 11 give

$$D_p = -(2p+3)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\mathcal{E}^2], \quad (26)$$

which Lemma 12 rewrites in the matrix form of Theorem 2.

Lemma 12 (Equivalent matrix form). *With K , S , S_γ , $\mathcal{E}$ and $\mathcal{E}_W$ as in Theorem 2,*

$$\begin{aligned} G &= \|\gamma\|^2, & F &= \gamma^\top K \gamma, & Q &= (\gamma^\top S)^2, & \mathcal{R} &= \|S_\gamma\|_F^2, \\ A_P &= \text{tr}(S_\gamma \mathcal{E}_W), & B_P &= (\gamma^\top S) \text{tr}(\mathcal{E}_W), & C_P &= \gamma^\top \mathcal{E}_W S. \end{aligned}$$

Proof. First,

$$F = \gamma_i \gamma_j T_{4,ijkl} \delta_{kl} = \gamma^\top \mathbb{E}[ZZ^\top \|Z\|^2] \gamma = \gamma^\top K \gamma,$$

and

$$S = \mathbb{E}[Z_i Z_j Z_j]_{i=1}^p = q$$

gives

$$Q = (\gamma^\top q)^2 = (\gamma^\top S)^2.$$

Next,

$$(S_\gamma)_{ab} = \mathbb{E}[Z_a Z_b (Z^\top \gamma)] = T_{3,abk} \gamma_k,$$

so

$$\|S_\gamma\|_F^2 = \sum_{a,b} (S_\gamma)_{ab}^2 = \gamma_k \gamma_\ell T_{3,abk} T_{3,abl} = \gamma_k \gamma_\ell L_{k\ell} = \mathcal{R}.$$

Since S_γ and $\mathcal{E}_W$ are symmetric,

$$\text{tr}(S_\gamma \mathcal{E}_W) = (S_\gamma)_{jk} (\mathcal{E}_W)_{jk} = \gamma_i T_{3,ijk} P_{jk} = A_P;$$

finally,

$$B_P = (\gamma^\top q) \text{tr}(P) = (\gamma^\top S) \text{tr}(\mathcal{E}_W)$$

and

$$C_P = \gamma^\top P q = \gamma^\top \mathcal{E}_W S.$$

$\square$

We close with an exact identity that is used by the variance-reduced verification of Appendix B.

Lemma 13 (Exact difference identity). *Let $\bar{X}_w$ and*

$$\mathcal{S}_w = \sum_{i:W_i=w} (X_i - \bar{X}_w)(X_i - \bar{X}_w)^\top$$

denote the within-arm means and scatter matrices, let

$$\hat{\beta}_w = \mathcal{S}_w^{-1} \sum_{i:W_i=w} (X_i - \bar{X}_w)(Y_i - \bar{Y}_w),$$

and write $d_X = \bar{X}_1 - \bar{X}_0$. Then, whenever $\mathcal{S}_0, \mathcal{S}_1$ and $\mathcal{S}_0 + \mathcal{S}_1$ are invertible and $n_1 = n_0$,

$$\hat{\tau}_{\text{REG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1} (\mathcal{S}_1 \hat{\beta}_1 + \mathcal{S}_0 \hat{\beta}_0),$$

$$\hat{\tau}_{\text{IREG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top \frac{\hat{\beta}_1 + \hat{\beta}_0}{2},$$

and consequently

$$\hat{\tau}_{\text{IREG}} - \hat{\tau}_{\text{REG}} = \frac{1}{2} d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1} (\mathcal{S}_1 - \mathcal{S}_0) (\hat{\beta}_1 - \hat{\beta}_0) = O_P(n^{-1}).$$

Proof. Both expressions for the estimators are the Frisch–Waugh–Lovell representations of (1): the coefficient on X in the REG regression is the pooled within-arm slope $(\mathcal{S}_1 + \mathcal{S}_0)^{-1} (\mathcal{S}_1 \hat{\beta}_1 + \mathcal{S}_0 \hat{\beta}_0)$, and IREG fits the two arms separately, so with $n_1 = n_0$ the overall covariate mean satisfies

$$\bar{X}_1 - \bar{X} = d_X/2 = -(\bar{X}_0 - \bar{X}).$$

Subtracting,

$$\begin{aligned} \hat{\tau}_{\text{IREG}} - \hat{\tau}_{\text{REG}} &= d_X^\top \left[(\mathcal{S}_1 + \mathcal{S}_0)^{-1} (\mathcal{S}_1 \hat{\beta}_1 + \mathcal{S}_0 \hat{\beta}_0) - \frac{1}{2} (\hat{\beta}_1 + \hat{\beta}_0) \right] \\ &= \frac{1}{2} d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1} (\mathcal{S}_1 - \mathcal{S}_0) (\hat{\beta}_1 - \hat{\beta}_0). \end{aligned}$$

The stochastic order follows from

$$d_X = O_P(n^{-1/2}), \quad (\mathcal{S}_1 + \mathcal{S}_0)^{-1} (\mathcal{S}_1 - \mathcal{S}_0) = O_P(n^{-1/2}), \quad \hat{\beta}_1 - \hat{\beta}_0 = O_P(1).$$

□

In whitened coordinates the leading term of this difference is

$$\frac{1}{4} d_Z^\top (A_1 - A_0) \gamma,$$

with A_w the arm- w mean of $ZZ^\top - I_p$; this is the control variate used in (27) below.

Lemma 14 (Leading expansion bias). *Under Assumption 1 and balanced complete randomization,*

$$\begin{aligned} \mathbb{E}[\tilde{\tau}_{n,\text{IREG}} - \tau] &= -\frac{4 \text{tr}(\mathcal{E}_W)}{n} + O(n^{-2}), \\ \mathbb{E}[\tilde{\tau}_{n,\text{REG}} - \tau] &= -\frac{4 \text{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}). \end{aligned}$$

Proof. Work in whitened coordinates with $m = n/2$ units per arm, arm slopes $b_1 - b_0 = \gamma$, projection residuals $\varepsilon_i(w)$ and within-arm scatter matrices $\mathcal{S}_w$. By Remark 3, the expansion bias is the mean of the quadratic part of $\hat{\tau} - \tau$, up to $O(n^{-2})$. For IREG,

$$\bar{Z}_1 - \bar{Z} = -(\bar{Z}_0 - \bar{Z}) = \frac{1}{2}(\bar{Z}_1 - \bar{Z}_0),$$

so

$$\hat{\tau}_{\text{IREG}} - \tau = \bar{\varepsilon}_1 - \bar{\varepsilon}_0 + \gamma^\top \bar{Z} - \frac{1}{2}(\bar{Z}_1 - \bar{Z}_0)^\top \{(\hat{b}_1 - b_1) + (\hat{b}_0 - b_0)\},$$

and the linear part of $\hat{b}_w - b_w$ is

$$m^{-1} \sum_{i: W_i = w} Z_i \varepsilon_i(w),$$

which enters the quadratic part of $\hat{\tau}_{\text{IREG}} - \tau$ through its product with $\bar{Z}_1 - \bar{Z}_0$. The arms are independent (Lemma 7), so

$$\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b}_1 - b_1)] = m^{-1} \mathbb{E}[\|Z\|^2 \varepsilon(1)] + O(n^{-2})$$

and

$$\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b}_0 - b_0)] = -m^{-1} \mathbb{E}[\|Z\|^2 \varepsilon(0)] + O(n^{-2}),$$

which gives

$$-(2m)^{-1} \text{tr}\{\mathbb{E}[ZZ^\top \varepsilon(1)] - \mathbb{E}[ZZ^\top \varepsilon(0)]\} = -\frac{4 \text{tr}(\mathcal{E}_W)}{n}.$$

For REG, the pooled slope $\hat{b}$ estimates $\bar{b} = (b_1 + b_0)/2$ and

$$\hat{\tau}_{\text{REG}} - \tau = \bar{\varepsilon}_1 - \bar{\varepsilon}_0 + \frac{1}{2} \gamma^\top (\bar{Z}_1 + \bar{Z}_0) - (\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b} - \bar{b}).$$

The pooled residual of unit i is $\varepsilon_i(W_i) + (W_i - \frac{1}{2})\gamma^\top Z_i$ up to centering, so the linear part of $\hat{b} - \bar{b}$ is

$$n^{-1} \sum_i Z_i \{\varepsilon_i(W_i) + (W_i - \frac{1}{2})\gamma^\top Z_i\},$$

which again enters the quadratic part of $\hat{\tau}_{\text{REG}} - \tau$ through its product with $\bar{Z}_1 - \bar{Z}_0$, and

$$\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top (\hat{b} - \bar{b})] = n^{-1} \{ \text{tr} \mathbb{E}[ZZ^\top \varepsilon(1)] - \text{tr} \mathbb{E}[ZZ^\top \varepsilon(0)] + \gamma^\top S \} + O(n^{-2}),$$

which gives the second display. $\square$

B Numerical verification of Theorem 2

The symbolic certificate verifies the algebra of Theorem 11 exactly. The role of simulation is different: it checks that the constant D_p governs the sampling behavior of the estimators, that the CRT correction in Theorem 10 is correct, and that no terms were lost in the derivation. Simulation cannot prove the theorem, so we use several checks that are designed to distinguish the stated formula from plausible alternatives.

Exact targets. We use data-generating processes with whitened covariates having independent coordinates with known moments. The oracle residuals take the form

$$\varepsilon(w) = g_w(Z) + s_w(Z)\eta,$$

where η is independent of Z with mean zero and unit variance, g_w is a polynomial orthogonal to $(1, Z)$, and s_w^2 is a polynomial. Each term in D_p is therefore a finite combination of population moments and can be computed in exact rational arithmetic. The code also checks $\mathbb{E}[g_w] = \mathbb{E}[Zg_w] = 0$, so that violations of the orthogonality conditions cannot pass unnoticed.

We use six designs, chosen so that every nontrivial block of D_p is active in at least one design: Gaussian homoskedastic covariates; centered-exponential covariates with heterogeneity; centered-exponential covariates with heteroskedasticity and $\gamma = 0$; centered-exponential covariates with heterogeneity, arm-specific misspecification, and arm-specific heteroskedasticity; Gaussian covariates with arm-specific misspecification and heteroskedasticity; and a mixed skewed/Gaussian design. These designs are analytically tractable and differ from those used in Section 3.3.

A variance-reduced estimator of D_p . The natural statistic $n^2\{\widehat{\text{MSE}}_{\text{IREG}} - \widehat{\text{MSE}}_{\text{REG}}\}$ has an unfavorable variance rate. By Lemma 13, the leading term of $\hat{\tau}_{\text{IREG}} - \hat{\tau}_{\text{REG}}$ is

$$d_1 = \frac{1}{4}d_Z^\top(A_1 - A_0)\gamma,$$

which is $O_P(n^{-1})$, while the common first-order error is

$$L = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0 = O_P(n^{-1/2}).$$

Thus

$$e_{\text{IREG}}^2 - e_{\text{REG}}^2 = 2d_1L + O_P(n^{-2}),$$

so the leading term of the scaled statistic is $2n^2d_1L = O_P(\sqrt{n})$. Under Assumption 1, this leading term has variance $O(n)$, so the corresponding per-replication standard deviation is $O(\sqrt{n})$; in the Gaussian design below, it increases with n as expected.

We remove this leading fluctuation with a control variate whose finite- n expectation is known exactly. Let $d_Z = \bar{Z}_1 - \bar{Z}_0$, let A_w be the arm- w mean of $ZZ^\top - I_p$, and define

$$d_1 = \frac{1}{4}d_Z^\top(A_1 - A_0)\gamma, \quad L = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0.$$

For an arm of size $m = n/2$, the expectation of a product of three centered IID sample means is the corresponding third joint cumulant divided by m^2 , giving the exact identity

$$\mathbb{E}[d_1L] = \frac{1}{n^2} \mathbb{E}\left[(\|Z\|^2 - 1)(\gamma^\top Z)\{\gamma^\top Z + \varepsilon(1) - \varepsilon(0)\}\right]. \quad (27)$$

Hence

$$\hat{D}_p = \frac{1}{B} \sum_{b=1}^B \left[n^2(e_{\text{IREG},b}^2 - e_{\text{REG},b}^2) - 2n^2d_{1,b}L_b \right] + 2n^2\mathbb{E}[d_1L]$$

is unbiased for the same finite- n scaled MSE difference as the raw statistic. Moreover,

$$n^2\{e_{\text{IREG}}^2 - e_{\text{REG}}^2 - 2d_1L\} = O_P(1),$$

so its per-replication variance is $O(1)$ rather than $O(n)$.

Both statistics are reported by the replication code. In the Gaussian design, the raw per-replication standard deviation increases from 151 at $n = 200$ to 293 at $n = 800$, while the control-variate standard deviation is 56 and 55, respectively. At $n = 3200$, the control-variate Monte Carlo standard error is ten times smaller than that of the raw statistic.

Results. Table 5 reports the exact targets and the control-variate estimates at $n \in \{200, 800, 3200\}$, using $B = 400,000$ replications for the two smaller sample sizes and $B = 120,000$ for $n = 3200$. The control variate keeps the per-replication standard deviation bounded, making the Monte Carlo errors small enough to resolve the finite- n remainder.

The largest discrepancy occurs in the all-blocks design (fourth row). There the estimate is 32.40 at $n = 200$, compared with the target 44.52, a difference of 12.1 (about 22 standard errors). The difference falls to 5.7 at $n = 800$ and 2.55 at $n = 3200$. The Bernoulli constant $D_{\text{Bern}} = D_p - 4G = 41.32$ is within one standard error of the $n = 3200$ estimate, so that sample size alone does not distinguish D_{Bern} from D_p in this design. The successive increments, 6.4 and 3.1, roughly halve as n quadruples, giving an extrapolated value of about 45, close to D_p . The ablation in Table 6 separates the two more clearly in other designs.

At $n = 3200$, all six estimates are within 2.6 standard errors of their targets. The two largest deviations are $z = -2.6$ in the all-blocks design and $z = +2.3$ in the $\gamma = 0$ heteroskedastic design. In the latter, the estimate moves from 29.82 at $n = 200$ and 31.73 at $n = 800$ to 33.25 at $n = 3200$. With only six designs, deviations of this size are compatible with Monte Carlo error. The results are consistent with a remainder that is smaller than the n^{-2} leading term, but three sample sizes are not enough to estimate its rate.

Table 5: Verification of Theorem 2. Targets are exact; estimates are the control-variate estimator with Monte Carlo standard errors in parentheses. $B = 400,000$ for $n \in \{200, 800\}$ and $B = 120,000$ for $n = 3200$.

DGP (active blocks)	D_p (exact)	$n = 200$	$n = 800$	$n = 3200$
Gaussian, homoskedastic, $p = 3$ (F)	-5.00	-4.70 (0.09)	-4.99 (0.09)	-5.17 (0.16)
Exponential covariates, homoskedastic (F, Q, R)	-20.25	-17.17 (0.21)	-19.60 (0.22)	-19.75 (0.39)
Exponential, heteroskedastic, $\gamma = 0$	32.00	29.82 (0.28)	31.73 (0.29)	33.25 (0.54)
Exponential, HTE + arm-specific misspec. (all)	44.52	32.40 (0.56)	38.82 (0.55)	41.97 (0.98)
Gaussian, arm-specific misspec. ($\text{tr } E^2, E_W$)	-2.32	-2.36 (0.09)	-2.34 (0.09)	-2.45 (0.16)
Mixed skewed/Gaussian, heteroskedastic	28.00	25.94 (0.27)	27.85 (0.28)	28.27 (0.53)

Power: what the verification would reject. A useful verification should be able to distinguish the stated formula from plausible alternatives. Table 6 reports z -statistics at $n = 3200$ for the exact D_p and several perturbed formulas: the Bernoulli constant $D_{\text{Bern}} = D_p - 4G$, which omits the CRT correction; the Gaussian special case $-(p+1)G$; and versions of D_p with individual blocks removed.

The exact formula has $|z| \leq 2.6$ in every design, whereas each alternative has $|z| > 3$ in at least one design, in some cases by tens of standard errors. The check has limited power when an alternative differs from D_p by less than the residual finite- n bias. In the all-blocks design, for example, $D_{\text{Bern}} = D_p - 3.2$ is not separated from D_p at $n = 3200$, which is why multiple designs are useful.

Without the control variate, the standard errors at $n = 3200$ would be 1.5 to 3.8, four to ten times larger in the heterogeneous designs. The Bernoulli constant would then not be separated from D_p in any design ($|z| \leq 3$ throughout), so the simulation would have little power to detect an omitted CRT correction.

Further covariate families and dimensions. The six designs above have $p \leq 3$ and independent coordinates. We also consider four covariate families at $p \in \{2, 5, 8\}$, including dependent coordinates (Table 7).

Table 6: Ablation: $z = (\hat{D}_p - D^{\text{alt}})/\widehat{\text{SE}}$ at $n = 3200$ for each design and candidate constant. Values with $|z| > 3$ indicate rejection of that candidate.

DGP	D_p	$D_p - 4G$	$-(p+1)G$	drop $-3\mathcal{R}$	drop $-Q$	drop resid.	drop $\mathcal{E}_W$
Gaussian, homoskedastic, $p = 3$ (F)	-1.0	+29.8	-1.0	-1.0	-1.0	-1.0	-1.0
Exponential covariates, homoskedastic (F, Q, R)	+1.3	+14.2	-41.3	-37.4	-21.9	+1.3	+1.3
Exponential, heteroskedastic, $\gamma = 0$	+2.3	+2.3	+61.1	+2.3	+2.3	+61.1	+2.3
Exponential, HTE + arm-specific misspec. (all)	-2.6	+0.7	+45.1	-12.4	-3.2	+65.8	-17.7
Gaussian, arm-specific misspec. ($\text{tr } E^2, E_W$)	-0.8	+23.5	+3.4	-0.8	-0.8	+3.4	-0.8
Mixed skewed/Gaussian, heteroskedastic	+0.5	+4.2	+55.7	-5.1	-1.4	+60.3	+0.5

For Gaussian covariates, $D_p = -(p+1)G$. For Rademacher covariates, $K = pI_p$ and $D_p = -(p+3)G$. For a symmetric scale mixture of normals, where 2% of units have six times the standard deviation,

$$K = (p+2)\tilde{\kappa}I_p, \quad \tilde{\kappa} = \frac{\mathbb{E}[s^4]}{\mathbb{E}[s^2]^2},$$

and

$$D_p = G\{(p+2)\tilde{\kappa} - (2p+3)\}.$$

For Gaussian covariates with quadratic misspecification $c(X^\top X - \text{tr } \Sigma)$, where $c = 0.3$ and $\Sigma_{jk} = 0.3^{|j-k|}$,

$$\mathcal{E} = 2c\Sigma, \quad D_p = -(p+1)G + 32c^2 \text{tr}(\Sigma^2).$$

Every family uses homoskedastic $N(0, 1)$ noise shared by the potential outcomes, with $G = 0.49$ (1.96 for the Rademacher family). Because both estimators are invariant to invertible linear transformations of the covariates, we simulate whitened covariates; dependence enters through γ and, for the misspecified family, through $\text{tr}(\Sigma^2)$.

The targets are exact. Recomputing them from the matrix form of Theorem 2 using 10^6 Monte Carlo draws reproduces them to within 5%, with the largest discrepancy in the scale mixture, where fourth moments are harder to estimate. Using exact targets avoids this additional Monte Carlo error.

With $n = 4,000$ and $B = 20,000$ replications per design, every estimate is within 1.55 standard errors of its target, and the control variate reduces the standard errors by factors of 3.9 to 11.4 relative to the raw estimator on the same replications.

Figure 2 shows the z -statistics for all 18 checks: the designs of Table 5 at $n = 3200$ and those of Table 7. Of these, 16 are within ± 1.96 ; the two exceptions are among the deviations discussed above.

Figure 3 shows the estimates as a function of n . For the four families at $p = 5$, the standard errors remain roughly constant as n grows. One of the twelve points below $n = 4,000$ lies outside ± 1.96 standard errors (the scale mixture at $n = 1,000$, $z = -2.21$), while the neighboring points at $n = 500$ and $n = 2,000$ are within 0.6 standard errors. With twenty-four family checks, an isolated deviation of this size is compatible with Monte Carlo variation.

The CRT correction and the bias decomposition. We check Theorem 10 separately by estimating the n^{-2} constant under both balanced CRT and Bernoulli(1/2) assignment on the same designs (`verification/crt_correction_certificate.py`). In the Gaussian design, the estimated gap between the two designs is 4.3 ± 0.6 at $n = 200$ and 6.3 ± 1.6 at $n = 800$, compared with the predicted $4G = 5$. In the exponential-covariate design, the corresponding gaps are 3.0 ± 1.1 and 8.6 ± 3.0 . All four comparisons have $|z| < 2$.

Table 7: Further verification of Theorem 2: four covariate families at $p \in \{2, 5, 8\}$, $n = 4,000$, $B = 20,000$. Targets are exact; estimates are the control-variate estimator with Monte Carlo standard errors in parentheses; the last column is the standard error of the raw estimator on the same replications.

Family	p	D_p (exact)	Estimate	z	Raw SE
Gaussian	2	-1.47	-1.31 (0.20)	+0.82	2.27
Gaussian	5	-2.94	-2.57 (0.42)	+0.90	3.22
Gaussian	8	-4.41	-3.50 (0.62)	+1.48	4.00
Scale mixture	2	14.81	13.33 (1.06)	-1.40	8.14
Scale mixture	5	25.56	26.51 (1.83)	+0.52	10.39
Scale mixture	8	36.30	35.37 (2.74)	-0.34	12.85
Rademacher	2	-9.80	-9.51 (0.29)	+1.02	2.97
Rademacher	5	-15.68	-16.65 (0.62)	-1.55	6.20
Rademacher	8	-21.56	-20.94 (0.94)	+0.67	8.13
Quadratic misspec.	2	4.81	4.75 (0.35)	-0.17	3.12
Quadratic misspec.	5	13.68	14.10 (0.96)	+0.44	4.99
Quadratic misspec.	8	22.56	22.19 (1.72)	-0.21	6.77

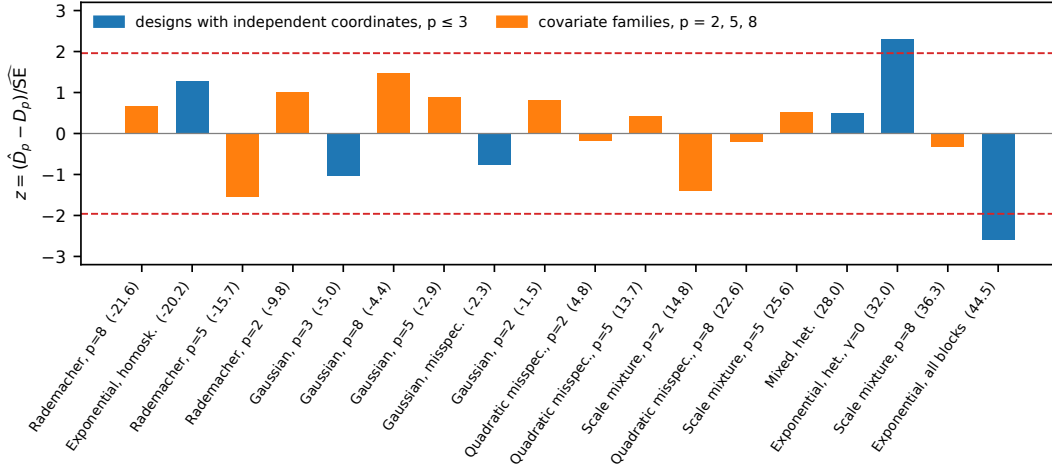


Figure 2: z -statistics $(\hat{D}_p - D_p) / \widehat{SE}$ of the control-variate estimates for the designs of Table 5 ($n = 3,200$) and Table 7 ($n = 4,000$), sorted by D_p (in parentheses). Dashed lines mark ± 1.96 .

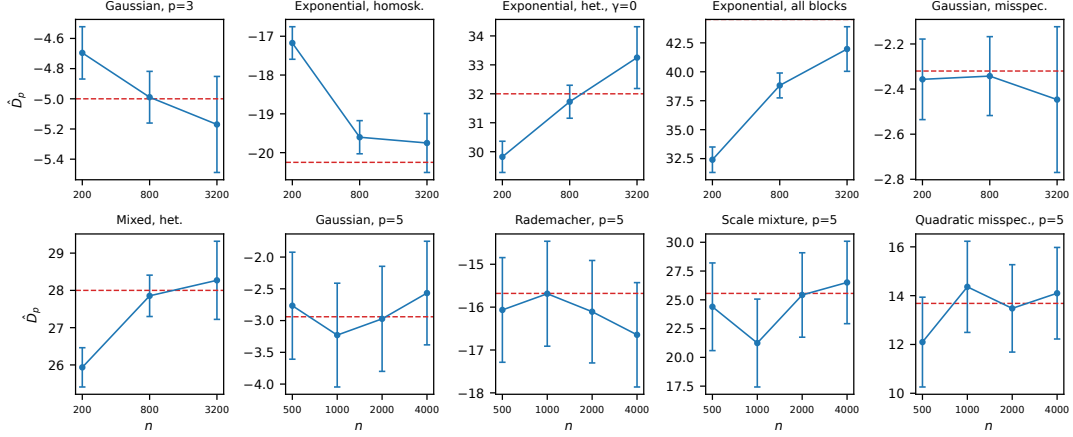


Figure 3: Control-variate estimates of D_p against n (log scale) with ± 1.96 standard-error bars; the dashed line is the exact D_p . The first six panels are the designs of Table 5 ($n = 200, 800, 3,200$); the last four are the families of Table 7 at $p = 5$ ($n = 500$ to $4,000$).

This check uses the raw statistic because the control variate is derived for the two-independent-samples representation of the CRT, so its power is limited. The ablation in Table 6 provides a sharper test: there, the Bernoulli constant is rejected by as many as thirty standard errors.

We also check Remark 3 by estimating the biases of both estimators directly (`simulations/run_bias.py`). With 10^6 replications, at $n = 800$ all six estimated biases (REG and IREG in three designs) are within 0.8 Monte Carlo standard errors of their leading-order predictions. At $n = 200$, the all-blocks design differs by about 11% of the predicted REG bias ($z = 9.3$); by $n = 800$, the difference is negligible at the reported Monte Carlo precision ($z = 0.4$), consistent with the $O(n^{-2})$ remainder in Lemma 14.

C Range of validity of the second-order approximation

Theorem 2 is an asymptotic result. A separate question is when the approximation

$$1 + \Delta/n$$

is accurate for the finite-sample MSE ratio. We examine this in the scalar lognormal family of the introduction, with $\sigma(X) = 1 + X$, now with a treatment-effect slope of 0.4 fixed while $X \sim \text{LogNormal}(0, s^2)$ varies. As s increases, the covariate tails become heavier and the higher moments entering Δ grow rapidly.

For $s = 0.8$ (skewness 3.7), the simulated excess MSE is 75% of the predicted excess Δ/n at $n = 500$ and 85% at $n = 2,000$; for $s = 0.6$, the corresponding figures are 91% and 94%. The approximation deteriorates as the tails become heavier. At $s = 1.5$, for example, the predicted ratio is four times the simulated ratio at $n = 500$. We do not interpret this as a universal threshold: the point is that a large deficiency can coexist with substantial finite-sample error in the second-order approximation.

The heavy-tailed example illustrates the same phenomenon more sharply. With $s = 1.8$, $\sigma(X) = 1 + 1.5X$, and treatment-effect slope 0.3, we have $\Delta \approx 3.9 \times 10^4$. The second-order

Table 8: Simulated MSE ratio $\widehat{\text{MSE}}_{\text{IREG}}/\widehat{\text{MSE}}_{\text{REG}}$ versus the second-order prediction $1 + \Delta/n$, for $X \sim \text{LogNormal}(0, s^2)$, $\sigma(X) = 1 + X$, treatment-effect slope 0.4. Monte Carlo standard errors of the simulated ratios are given in parentheses in units of the last reported digit. The introductory example is this family at $s = 1.1$ with no treatment-effect heterogeneity; the heavy-tailed example ($s = 1.8$, $\sigma(X) = 1 + 1.5X$, slope 0.3) is a separate design.

s	skew Z	Δ	$n = 500$		$n = 2000$		$n = 8000$	
			sim.	pred.	sim.	pred.	sim.	pred.
0.6	2.3	4	1.0072 (1)	1.0079	1.0019 (1)	1.0020	1.0004 (1)	1.0005
0.8	3.7	12	1.0184 (3)	1.0243	1.0052 (2)	1.0061	1.0014 (1)	1.0015
1.0	6.2	41	1.0429 (7)	1.0827	1.0137 (4)	1.0207	1.0043 (2)	1.0052
1.2	11.2	166	1.0923 (15)	1.3323	1.0347 (11)	1.0831	1.0127 (8)	1.0208
1.5	33.5	2025	1.2517 (51)	5.0503	1.1163 (44)	2.0126	1.0534 (36)	1.2531

prediction is then 78.5, 20.4, and 5.8 at $n = 500$, 2,000, and 8,000, compared with simulated MSE ratios of 1.68, 1.34, and 1.19. The sign of the comparison is correct, but $1 + \Delta/n$ is no longer a useful numerical approximation to the MSE ratio. In the same design, $R_\tau^2 \approx 0.009$, and the simulated coverage of IREG is 85.6% with HC1, 88.9% with HC2, and 94.0% with HC3, compared with 94.4% for REG+HC1 and 96.7% for REG+HC3.

Tail sensitivity. The largest discrepancies occur when the moments entering Δ are dominated by the far tail. Table 9 recomputes $1 + \Delta/n$ at $n = 500$ after truncating the lognormal law at several high quantiles; the truncated moments are available in closed form. Truncating at the $1 - 10^{-4}$ quantile reduces Δ by one to three orders of magnitude, showing how strongly the deficiency depends on extreme covariate values. Even truncation at the $1 - 10^{-8}$ quantile still gives predictions several times larger than the simulated ratios.

In these designs, the problem begins already at first order. For $\gamma = 0$, at $s = 0.6$ the simulated standard deviation of $\hat{\tau}_{\text{REG}}$ is close to the first-order benchmark $\sqrt{V^*/n}$, 0.21 versus 0.21. At $s = 1.5$ and $s = 2$, the corresponding values are 0.72 versus 0.88 and 2.46 versus 4.90, respectively. Thus the heavy-tailed designs are outside the regime where the perturbative expansion gives an accurate quantitative approximation. Monte Carlo error is a secondary issue in the heaviest rows, where rare draws contribute substantially to the simulated MSE. The sign of the REG–IREG comparison agrees with the second-order calculation in every design.

Table 9: Second-order prediction $1 + \Delta/n$ at $n = 500$ for the full lognormal law and for its truncation at the q quantile, against the simulated MSE ratio (Monte Carlo standard error in parentheses, in units of the last digit).

Design	Simulated ratio	full law	$1 + \Delta/n$		
			$q = 1 - 10^{-4}$	$q = 1 - 10^{-6}$	$q = 1 - 10^{-8}$
Heavy-tailed example ($s = 1.8$)	1.68 (3)	78.5	1.90	7.94	26.6
Table 8, $s = 1.5$	1.25 (1)	5.05	1.36	2.44	3.81
Section 3.3, $\gamma = 0$, $s = 2$	2.00 (6)	672.1	2.45	17.3	97.4

D Implementation of the inference simulations

The simulations of Section 3.3 use `estimatr` 1.0.2. For speed, each replication calls `estimatr`'s internal fitting routine, the function behind `lm_robust` and `lm_lin`, with the design matrices those functions construct; on 100 simulated datasets this reproduces their estimates, standard errors and degrees of freedom exactly. Because `estimatr`'s default without clustering is HC2, we request HC1 and HC3 explicitly, and intervals use the degrees of freedom `estimatr` reports. The simulations of Sections 3.4 and 3.5 and part of Appendix E use a NumPy implementation of the same estimators, which agrees with `estimatr` and `statsmodels` to 4×10^{-14} . Further checks are documented in the replication repository.

Leverage-one handling. At a leverage-one observation the HC2 and HC3 contributions are 0/0, and software resolves this differently. `estimatr` 1.0.2 evaluates the ratio in floating point, so HC2 is frequently undefined (no interval in 53.6% of the leverage-one replications of the rare-value design with $\sigma = 1 + |X_{i1}|$) and HC3 receives a rounding-determined contribution; its 94.9% coverage there does not reflect a valid variance estimate. The development version of `estimatr` on GitHub (version 2.0.0, commit `e70f3ee`) drops the contribution of an observation whose computed leverage is at or above one; under it IREG+HC3 covers 90.5% and IREG+HC2 85.8% of those replications. Table 15 reports both versions.

E The superpopulation correction: a calibration diagnostic

This appendix separates the population-ATE variance component G/n from its plug-in estimate $\hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}/n$. On each simulated dataset we form five intervals using the same IREG estimate and degrees of freedom: HC1; HC1 plus the plug-in correction; HC1 plus the true G/n (an oracle); HC3; and HC3 plus the plug-in correction. Only the t critical value changes with the nominal level. When $G = 0$, the oracle and uncorrected intervals coincide.

The Gaussian designs used in the correction study have $X_i \sim N(0, I_p)$,

$$Y_i(0) = 0.9 \sum_j X_{ij} + u_i(0), \quad Y_i(1) = Y_i(0) - u_i(0) + u_i(1) + 1 + 0.3X_{i1},$$

with $N(0, 1)$ residuals. We consider five designs: residuals shared across arms at $(n, p) = (500, 5)$, $(1000, 2)$, $(1000, 5)$, and $(1000, 10)$, and independent residuals across arms at $(n, p) = (500, 5)$. The calibration diagnostic uses three designs, all with $n = 500$.

Large R_τ^2 : $p = 5$, $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$, $Y_i(0) = 0.9 \sum_j X_{ij} + u_i$, and $Y_i(1) = Y_i(0) + 1 + 3X_{i1}$, with $u_i \sim N(0, 1)$. Here $G = 9$ and $R_\tau^2 = 9/13$; there are 20,000 replications.

Small R_τ^2 : the heavy-tailed example, with $X \sim \text{LogNormal}(0, 1.8^2)$, $\sigma(X) = 1 + 1.5X$, and treatment-effect slope 0.3; there are 100,000 replications.

$\gamma = 0$: the $s = 2$ design from Table 1, with 20,000 replications.

The first design uses the NumPy implementation of Appendix D, the second uses the implementation for the scalar lognormal examples, and the third uses `estimatr` 1.0.2. Monte Carlo standard errors are at most 0.4 percentage points.

Table 10 gives coverage of the 95% intervals. Its last column reports the median ratio of the HC1-plus-plug-in interval width to the HC1 width. Table 11 and Figure 4 show coverage across nominal levels from 80% to 99%.

The simulations show three distinct regimes.

Table 10: Coverage (%) of nominal 95% intervals for the population ATE, and the median ratio of interval widths (HC1 plus plug-in over HC1).

Design	R_τ^2	HC1	HC1 + plug-in	HC1 + G/n (oracle)	Median width ratio
$\gamma = 0$, skewed ($s = 2$)	0	83.0	95.1	83.0	1.07
Small R_τ^2 , heavy-tailed example	0.009	85.6	94.4	86.5	1.06
Large R_τ^2	0.692	72.7	95.2	95.1	1.80

Table 11: Coverage (%) of intervals for the population ATE at several nominal levels. “Plug-in” adds $\hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}/n$ to the robust variance; “ G/n (oracle)” adds the true first-order component.

Method	80%	90%	95%	98%	99%
$\gamma = 0$, skewed ($s = 2$): $R_\tau^2 = 0$ exactly					
HC1	64.4	75.5	83.0	89.1	91.9
HC1 + plug-in	74.6	88.3	95.1	98.6	99.5
HC1 + G/n (oracle)	64.4	75.5	83.0	89.1	91.9
HC3	79.5	89.2	93.9	96.9	98.0
HC3 + plug-in	85.7	94.6	98.1	99.5	99.8
REG, HC1	76.3	88.8	95.3	98.6	99.5
Small R_τ^2 , heavy-tailed example: $R_\tau^2 = 0.009$					
HC1	67.2	78.5	85.6	91.2	93.7
HC1 + plug-in	75.3	88.1	94.4	97.9	99.1
HC1 + G/n (oracle)	68.2	79.5	86.5	91.8	94.2
HC3	79.8	89.3	94.0	97.0	98.2
HC3 + plug-in	84.9	93.6	97.3	99.1	99.6
REG, HC1	76.2	88.3	94.4	97.8	99.0
Large R_τ^2 : $R_\tau^2 = 0.692$					
HC1	52.1	63.9	72.7	80.7	85.3
HC1 + plug-in	80.2	90.0	95.2	98.1	99.1
HC1 + G/n (oracle)	80.2	90.0	95.1	98.2	99.2
HC3	52.7	64.6	73.3	81.3	85.7
HC3 + plug-in	80.4	90.1	95.3	98.2	99.1
REG, HC1	80.2	90.2	95.2	98.2	99.1

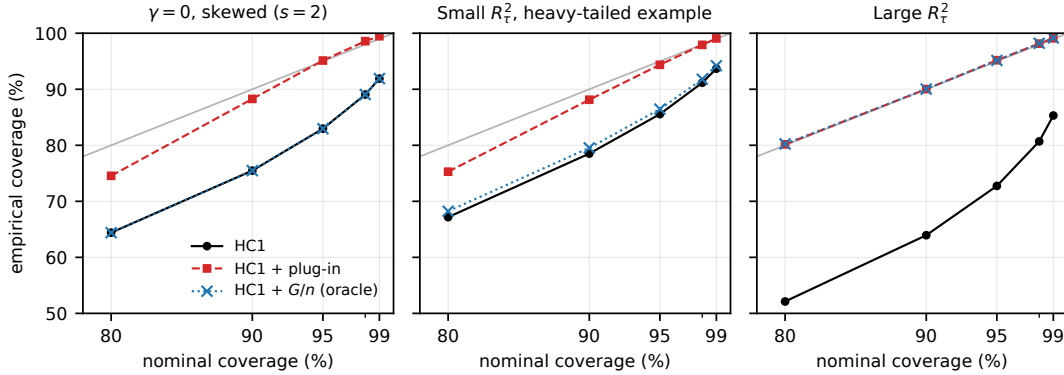


Figure 4: Empirical against nominal coverage for IREG with HC1, HC1 plus the plug-in correction, and HC1 plus the true G/n , in the three designs of Table 11. The grey line is the diagonal. With $\gamma = 0$ (left) the oracle coincides with HC1.

With large R_τ^2 , the plug-in estimates an average of 1.00 times G/n . The plug-in and oracle intervals have similar coverage across the nominal levels, while HC1 alone undercovers. For example, at the 80% level the two corrected methods give 80.2% and 80.2%, respectively; at 99% they give 99.1% and 99.2%. The median width ratio is 1.80.

In the small- R_τ^2 example, the oracle changes HC1 coverage by only about one percentage point, whereas the plug-in changes it by five to ten points depending on the nominal level. The same pattern appears when $\gamma = 0$: the oracle has no effect, but the plug-in still changes coverage substantially. The median width ratios in these two designs are only 1.06 and 1.07. Thus a modest average increase in interval width can have a large effect on coverage when the HC1 intervals are already poorly calibrated.

The width changes also depend on where the correction is active. With $\gamma = 0$, the median width ratio is 1.48 among replications where the HC1 interval misses and 1.05 where it covers. In the large- R_τ^2 design the corresponding ratios are 1.80 and 1.80. Interval width alone therefore does not determine calibration.

Across nominal levels, the plug-in remains poorly calibrated in the two small- R_τ^2 designs. In particular, it undercovers at 80% with coverage 75.3% and 74.6%, while at 99% it gives 99.1% and 99.5%. The agreement at 95% is therefore specific to that nominal level; REG+HC1 shows a similar, but smaller, pattern in the $\gamma = 0$ design.

HC3 alone is within about one percentage point of nominal across these levels in the $\gamma = 0$ design. Adding the plug-in correction to HC3 then produces overcoverage; for example, coverage is 85.7% at the 80% level. Because $G = 0$ in this design, this effect comes from the finite-sample behavior of the plug-in estimator rather than from the population-ATE variance component itself.

Results at the 95% level for the 20 designs in the paper and 19 supplementary designs are reported in `results/correction_summary.csv`.

F Supplementary results for Section 3

F.1 Further designs with $\gamma = 0$

The designs in Table 12 vary the misspecification, sample size, and number of covariates. They use

$$Y_i(0) = 0.9 \sum_j X_{ij} + g(X_i) + \sigma(X_i)u_i, \quad Y_i(1) = Y_i(0) + 1,$$

with $p = 5$, $X_{i1} \sim \text{LogNormal}(0, s^2)$, $X_{i2}, \dots, X_{i5} \sim N(0, 1)$, and $\sigma(X) = 1 + X_{i1}$. We use 20,000 replications (10,000 at $n = 2,000$ and 4,000 at $n = 8,000$), giving Monte Carlo standard errors of at most 0.36 percentage points.

With Gaussian covariates, quadratic misspecification has little effect. With a skewed covariate, misspecification that grows in the tail can produce substantial undercoverage even with homoskedastic noise (93.0% for IREG+HC1). In the base $s = 1.2$ design, IREG+HC1 coverage improves from 91.3% at $n = 200$ to 94.6% at $n = 2,000$ and 94.5% at $n = 8,000$. The pattern persists with $p = 5$.

Table 12: Coverage (%) of nominal 95% intervals when $\gamma = 0$ in further designs. Unless stated, $p = 1$ and $X \sim \text{LogNormal}(0, s^2)$; Z is the standardized covariate.

Design	n	Reps	IREG		REG		
			HC1	HC3	HC1	HC3	
<i>Misspecification, homoskedastic unless noted</i>							
Gaussian X , $0.3(X^2 - 1)$	500	20,000	94.9	95.0	94.9	95.0	
$s = 1$, $2 \log X$	500	20,000	94.5	95.7	95.0	95.6	
$s = 1$, $0.3(Z^2 - 1)$	500	20,000	93.0	95.7	95.1	96.2	
$s = 1$, $2 \log X$, $\sigma = 1 + X$	500	20,000	94.2	95.5	94.9	95.6	
<i>Sample size</i>							
$s = 1.2$, $\sigma = 1 + X$	200	20,000	91.3	94.9	94.8	96.4	
$s = 1.2$, $\sigma = 1 + X$	500	20,000	92.6	95.1	94.8	95.8	
$s = 1.2$, $\sigma = 1 + X$	2,000	10,000	94.6	95.6	95.4	95.8	
$s = 1.2$, $\sigma = 1 + X$	8,000	4,000	94.5	94.9	94.8	95.0	
<i>More covariates</i>							
$p = 5$, $s = 1$, $\sigma = 1 + X_1$	500	20,000	94.2	95.5	95.1	95.7	
$p = 5$, $s = 1$, $\sigma = 1 + X_1$	200	20,000	93.1	95.5	94.9	96.1	

F.2 Sampling variability and the second-order prediction

Table 13 compares the sampling variability of the two estimators with their reported standard errors. At $s = 2$, the Monte Carlo standard deviations are 3.48 for IREG and 2.46 for REG, giving an MSE ratio of 2.00. Their median HC1 standard errors are nearly identical, at 1.75 and 1.73. Thus in this design the larger variability of IREG is not reflected in its HC1 standard errors. The sampling distributions are heavy-tailed for both estimators, so Monte Carlo standard deviations also exceed their median reported standard errors.

The second-order prediction is accurate for moderate tails but deteriorates as s increases. At $s = 0.8$, $1 + \Delta/n = 1.03$ versus a simulated ratio of 1.02. For $s \geq 1$, the prediction increasingly overstates the finite-sample ratio, as discussed in Appendix C. The simulated ratio remains above one in every design, as predicted by the sign of Δ .

Table 13: Monte Carlo standard deviation and MSE of $\hat{\tau}_{\text{IREG}}$ and $\hat{\tau}_{\text{REG}}$ against their median HC1 and HC3 standard errors, in the designs of Table 1; $\sqrt{V^*/n}$ is the common first-order standard deviation of the two estimators and $1 + \Delta/n$ the second-order prediction of Theorem 2; Monte Carlo standard errors of the simulated MSE ratio are in parentheses, in units of the last digit.

s	Standard deviation			MSE ratio IREG/REG		Median standard error			
	IREG	REG	$\sqrt{V^*/n}$	simulated	$1 + \Delta/n$	IREG HC1	REG HC1	IREG HC3	
0.6	0.211	0.210	0.209	1.007 (<1)	1.01	0.207	0.207	0.210	
0.8	0.242	0.240	0.243	1.017 (1)	1.03	0.237	0.237	0.243	
1.0	0.301	0.295	0.306	1.043 (2)	1.09	0.288	0.287	0.301	
1.2	0.420	0.401	0.428	1.096 (3)	1.34	0.373	0.371	0.402	
1.5	0.812	0.721	0.882	1.271 (14)	4.81	0.618	0.614	0.714	
1.8	1.864	1.464	2.303	1.620 (39)	72	1.128	1.118	1.414	
2.0	3.482	2.465	4.896	1.996 (56)	672	1.748	1.731	2.327	

F.3 HC3 and interval width

HC3 largely removes the variance-scale error in the $s = 2$ design. Its implied scale is 1.05, and the ratio of empirical to t quantiles of $|T|$ rises from 1.01 at 80% to 1.14 at 99%, leaving some tail distortion beyond a simple scale adjustment.

Table 14 shows the cost of HC3. Along the skewness sweep, the median HC3/HC1 standard-error ratio for IREG rises from 1.01 at $s = 0.6$ to 1.28 at $s = 2$, with a 90th percentile of 1.92. For REG at $s = 2$, the corresponding values are 1.09 and 1.34. The adjustment is concentrated in a minority of replications: at $s = 2$ the median HC1 widths of IREG and REG are still close, at 6.87 and 6.80. HC3 remains somewhat below nominal under the most extreme tails (Table 1, $s \geq 1.8$).

Table 14: Median width of nominal 95% intervals and the ratio $\text{SE}_{\text{HC3}}(\hat{\tau})/\text{SE}_{\text{HC1}}(\hat{\tau})$ (median and 90th percentile over replications) in the designs of Table 1.

s	Median width				IREG HC3/HC1		REG HC3/HC1	
	IREG HC1	IREG HC3	REG HC1	REG HC3	median	p90	median	p90
0.6	0.82	0.82	0.81	0.82	1.01	1.02	1.01	1.01
0.8	0.93	0.95	0.93	0.94	1.02	1.05	1.01	1.03
1.0	1.13	1.18	1.13	1.15	1.04	1.11	1.02	1.05
1.2	1.47	1.58	1.46	1.51	1.07	1.22	1.03	1.10
1.5	2.43	2.81	2.41	2.57	1.13	1.45	1.05	1.19
1.8	4.43	5.56	4.39	4.81	1.22	1.71	1.07	1.28
2.0	6.87	9.14	6.80	7.57	1.28	1.92	1.09	1.34

F.4 Covariate representation

On the same $s = 2$ datasets, replacing X by $\log(1 + X)$ reduces the largest IREG leverage from 0.65 to 0.08 and raises IREG+HC1 coverage from 83.0% to 95.6%. The median interval width increases from 6.87 to 9.92 because the outcome is linear in X in this design. This is one illustration that the leverage problem depends on the chosen covariate representation.

F.5 Leverage one and software

An extreme covariate value can have leverage one within an arm. With one such observation, the arm-specific fit interpolates that observation, so its residual is zero. Residual-based variance estimators therefore cannot recover its contribution from that residual; HC2 and HC3 also encounter a 0/0 term.

We isolate this case with a rare-value design: $p = 1$, $n = 500$, $\rho = 0.02$, with

$$X_{i1} = \begin{cases} \sqrt{(1-\rho)/\rho}, & \text{with probability } \rho, \\ -\sqrt{\rho/(1-\rho)}, & \text{otherwise,} \end{cases}$$

and $\sigma = 1 + |X_{i1}|$ or $\sigma = 1$.

An arm contains exactly one rare unit, and the other arm contains at least one, in 6.1% of replications (exact probability 6.39%). In the heteroskedastic design, the isolated unit also has the largest residual variance. IREG+HC1 covers 85.6% of these replications. With $\sigma = 1$, coverage is 95.7%. Rank deficiency occurs in 1.4% of replications (exact probability 1.28%).

Software. The treatment of the 0/0 HC2/HC3 contribution at leverage one is software-dependent [Kranz, 2024]. Table 15 reports two versions of `estimatr`; they differ in how they handle this term and in which collinear column is dropped in rank-deficient fits. Neither recovers the variance contribution of the leverage-one observation.

Table 15: Coverage (%) in the rare-value design ($\gamma = 0$, $p = 1$, $n = 500$, $\rho = 0.02$, 20,000 replications). Rows: all replications as the software reports them; replications with a full-rank interacted design; and, within those, replications in which an arm contains a leverage-one unit. With $\sigma = 1 + |X_1|$ these are 1.4% rank-deficient and 6.1% leverage-one replications; with $\sigma = 1$, 1.3% and 6.6%; the exact probabilities are 1.28% and 6.39%. The two `estimatr` versions are applied to the same simulated datasets; they differ only in how they evaluate the 0/0 HC2/HC3 contribution and in which collinear column they drop in rank-deficient fits. REG does not depend on the version. A replication without a finite interval counts as a non-cover.

	IREG, v1.0.2			IREG, v2.0.0			REG	
	HC1	HC2	HC3	HC1	HC2	HC3	HC1	HC3
<i>Rare value, $\sigma = 1 + X_1$</i>								
all replications	92.0	90.1	94.8	92.0	93.1	94.6	94.9	96.0
full-rank designs	92.2	90.5	95.1	92.2	93.3	94.9	94.9	96.0
of which leverage-one	85.6	40.2	94.9	85.6	85.8	90.5	95.7	96.6
<i>Rare value, $\sigma = 1$</i>								
all replications	94.7	90.9	94.7	94.7	94.7	94.9	95.0	95.0
full-rank designs	94.9	91.2	95.2	94.9	94.9	95.1	94.9	95.0
of which leverage-one	95.7	41.3	97.6	95.7	95.7	97.1	95.8	95.9

F.6 Designs of the DGP selection

The DGP selection in Section 3.5 combines the scalar lognormal designs with the $p = 5$ designs from the correction study. The lognormal designs use

$$p = 1, \quad X_i \sim \text{LogNormal}(0, s^2), \quad Y_i(0) = 0.9X_i + \sigma(X_i)u_i, \quad Y_i(1) = Y_i(0) + 1 + gX_i,$$

with $\sigma(X) = 1 + X$ except in the $s = 1.8$ heavy-tailed design, where $\sigma(X) = 1 + 1.5X$, and

$$(s, g) \in \{(1.2, 0.3), (1.5, 0.1), (1.5, 0.5), (1.8, 0.3), (2, 0.1), (1.5, 1), (1.5, 3)\}.$$

The remaining designs have $p = 5$, $n = 500$, potential outcomes that share their noise, and

$$Y_i(0) = m(X_i) + \sigma(X_i)u_i, \quad Y_i(1) = Y_i(0) + 1 + \gamma^* X_{i1}, \quad u_i \sim N(0, 1),$$

with $m(x) = 0.9 \sum_j x_j$ unless stated. They differ in the covariate law, the noise scale and the slope γ^* :

- *Gaussian*: $X_i \sim N(0, I_5)$, $\sigma \equiv 1$ and $\gamma^* = 0.3$; the large-heterogeneity Gaussian row instead has $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$ and $\gamma^* = 3$.
- *Misspecified, $G = 0$* : $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$, $\sigma \equiv 1$, $\gamma^* = 0$, and $m(x) = 0.9 \sum_j x_j + 0.20(x_1^2 - 1) + 0.15x_1x_2$, so that the working model is misspecified in both arms.
- *Skewed, het., $G = 0$* : independent coordinates, the first a centered LogNormal($0, 1.1^2$) and the others standard normal, mixed by the Cholesky factor of the matrix $0.35^{|j-k|}$ and then standardized to unit variances; $\sigma(X) = 1 + 0.5|X_{i1}| + 0.25\|X_i\|$ and $\gamma^* = 0$.
- *Heavy-tailed, het.*: the same construction with LogNormal($0, 1.5^2$) and $\sigma(X) = 1 + 0.75|X_{i1}| + 0.25\|X_i\|$; $\gamma^* = 0.5$ in the small-heterogeneity row and $\gamma^* = 3$ in the large-heterogeneity row.
- *Rare extreme values*: X_{i1} is the two-valued covariate of the rare-value design above with $\rho = 0.02$, $X_{i2}, \dots, X_{i5} \sim N(0, 1)$, $\sigma \equiv 1$ and $\gamma^* = 1$.

The Gaussian designs are those of Appendix E; the others are used only here.

For the lognormal designs with small heterogeneity, IREG+HC3 without the plug-in correction has coverage between 79.0% and 80.7% at nominal 80%. Table 16 gives the corresponding 95% results for IREG with HC1, HC1 plus the oracle correction, and HC3, together with the ratio of the average plug-in term to G/n .

Table 16: Further intervals for the designs of Table 3: coverage (%) of nominal 95% intervals for the population ATE for IREG without the plug-in and with the true G/n , and the ratio of the average plug-in term to G/n .

Design	R_τ^2	IREG			Plug-in $/(G/n)$
		HC1	HC1 + oracle	HC3	
Gaussian	0.02	94.6	94.9	94.9	1.4
Misspecified, $G = 0$	0	94.8	94.8	95.1	—
Skewed, het., $G = 0$	0	94.6	94.6	95.6	—
Heavy-tailed, het.	0.02	93.2	93.5	95.3	5.7
Lognormal $s = 1.2$	0.01	92.6	92.9	94.9	6.7
Lognormal $s = 1.5$	0.002	90.3	90.4	94.9	71.0
Lognormal $s = 1.5$	0.05	88.8	90.5	93.8	4.1
Lognormal $s = 1.8$	0.009	85.5	86.4	94.0	25.6
Lognormal $s = 2$	0.002	83.1	83.5	93.8	90.2
Lognormal $s = 1.5$	0.17	85.3	91.9	91.2	1.7
Rare extreme values	0.20	90.8	93.8	91.2	1.0
Heavy-tailed, het.	0.37	86.8	94.6	89.6	1.1
Lognormal $s = 1.5$	0.65	64.4	95.2	72.2	1.1
Gaussian	0.69	72.2	95.1	72.9	1.0